EX. NO: 01 DATE:	**Use built-in functions to check the datatypes of variables and perform coercion**

AIM :

To Use built-in functions to check the datatypes of variables and perform coercion Using R Program.

ALGORITHM :

Step 1 : Create variables using assignment operator <-

Step 2 : Check datatype using class()

Step 3 : Convert datatype using coercion functions

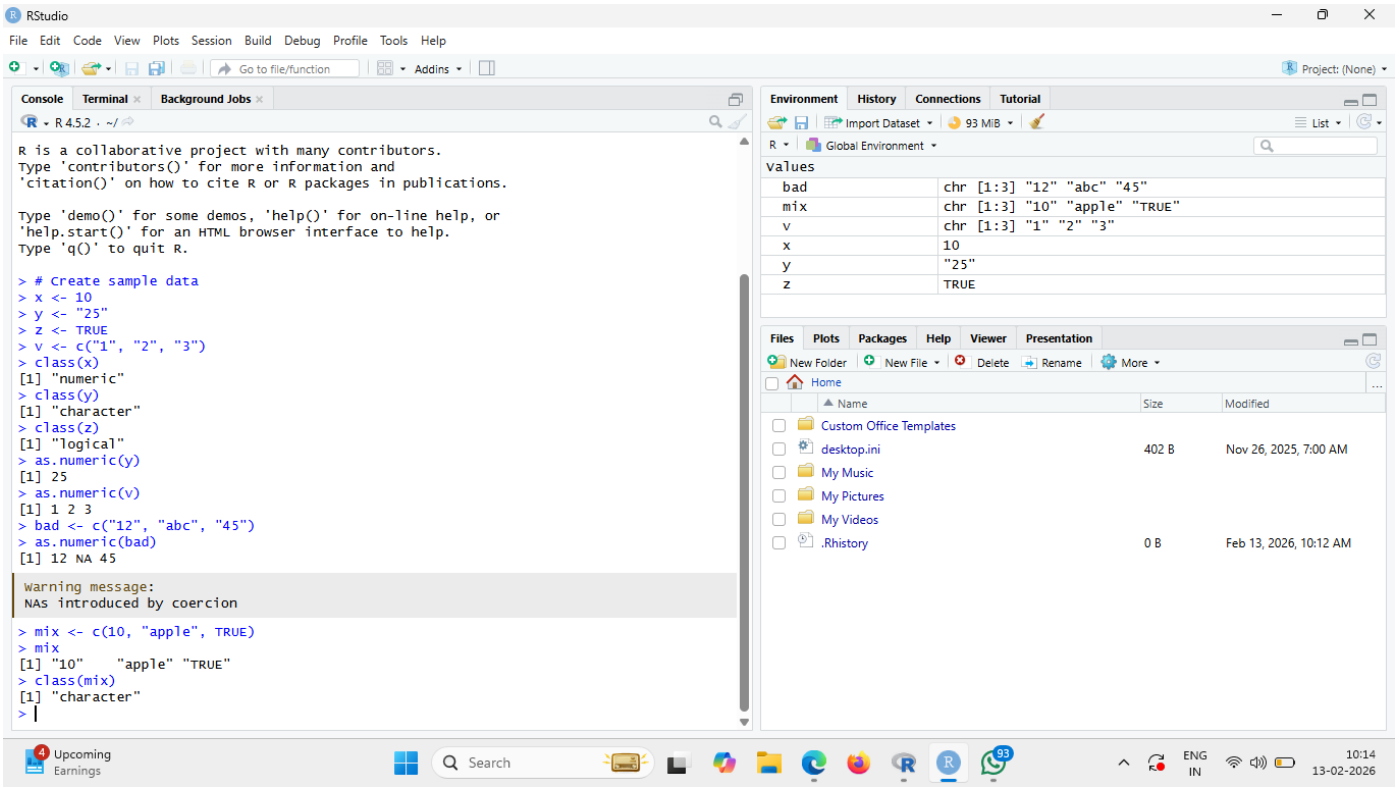
Step 4 : Handle invalid conversion Non-numeric values convert to NA

Step 5 : Understand automatic coercion rule When different types are combined in a vector, R converts all elements to one common type in the order: logical → numeric → character

SOURCE CODE:

```
# Create sample data
x <- 10
y <- "25"
z <- TRUE
 v <- c("1", "2", "3")
class(x)
class(y)
class(z)
as.numeric(y)
 as.numeric(v)
bad <- c("12", "abc", "45")
as.numeric(bad)
mix <- c(10, "apple", TRUE)
mix
class(mix)
```

OUTPUT:



RESULT:

The above program is executed successfully .

EX. NO: 02

DATE:

IMPLEMENT VECTOR OPERATIONS

AIM :

To Implement Vector Operations Using R Program .

ALGORITHM :

Step 1 : Start the program.

Step 2 : Create two vectors `a = c(10,20,30)` and `b = c(1,2,3)`.

Step 3 : Perform arithmetic operations (`+`, `-`, `*`, `/`) on `a` and `b`.

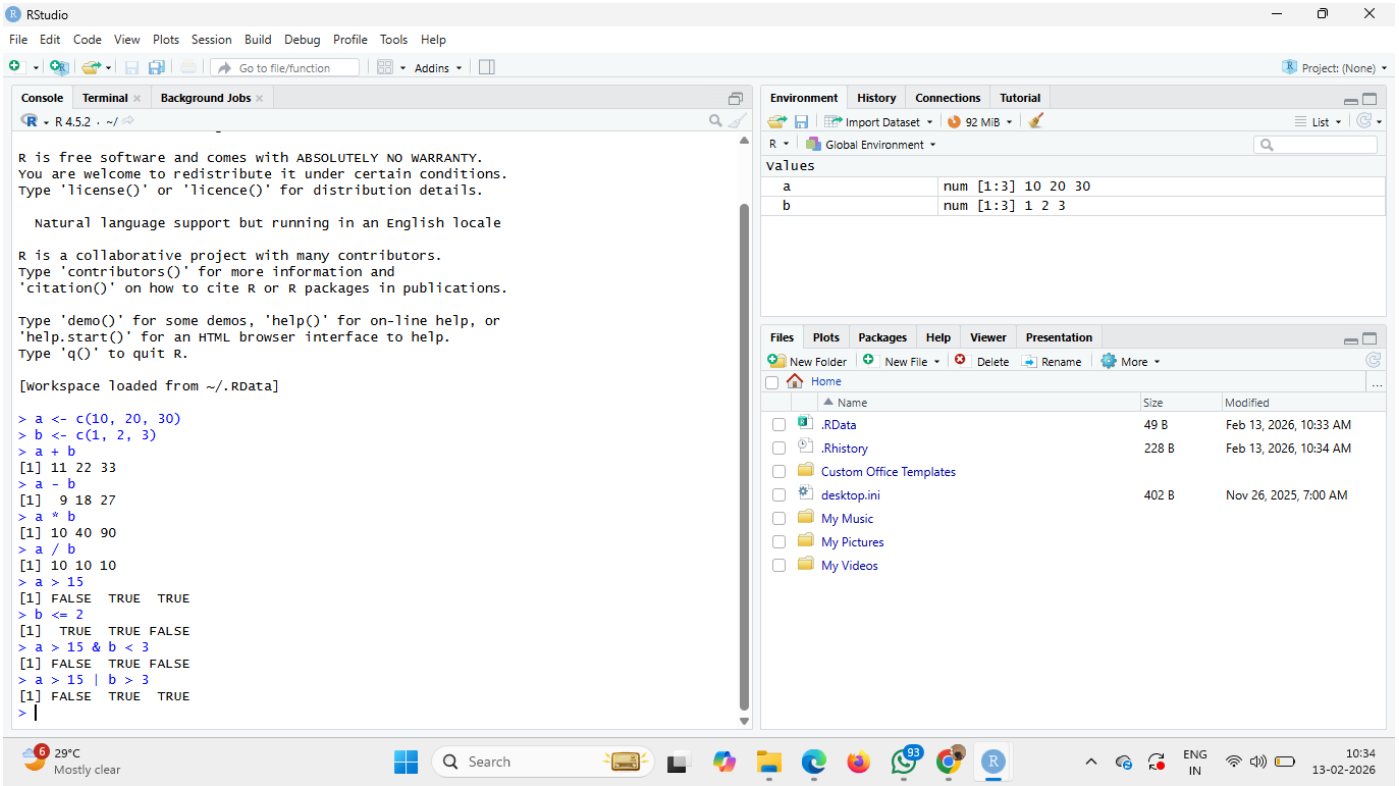
Step 4 : Perform relational and logical operations (`>`, `<=`, `&`, `|`).

Step 5 : Display the results and stop.

SOURCE CODE:

```
a <- c(10, 20, 30)
b <- c(1, 2, 3)
a + b
a - b
a * b
a / b
a > 15
b <= 2
a > 15 & b < 3
a > 15 | b > 3
```

OUTPUT:



RESULT:

The above program is executed successfully .

EX. NO: 03 DATE:	PERFORM MATRICS AND LIST OPERATIONS

AIM :

To Perform Matrics and List Operations Using R Program .

ALGORITHM :

Step 1: Start the program.

Step 2: Create a matrix `m ← matrix(1:9, nrow=3, byrow=TRUE)` and display it.

Step 3: Perform matrix operations: multiply `m * 5` and find transpose `t(m)`.

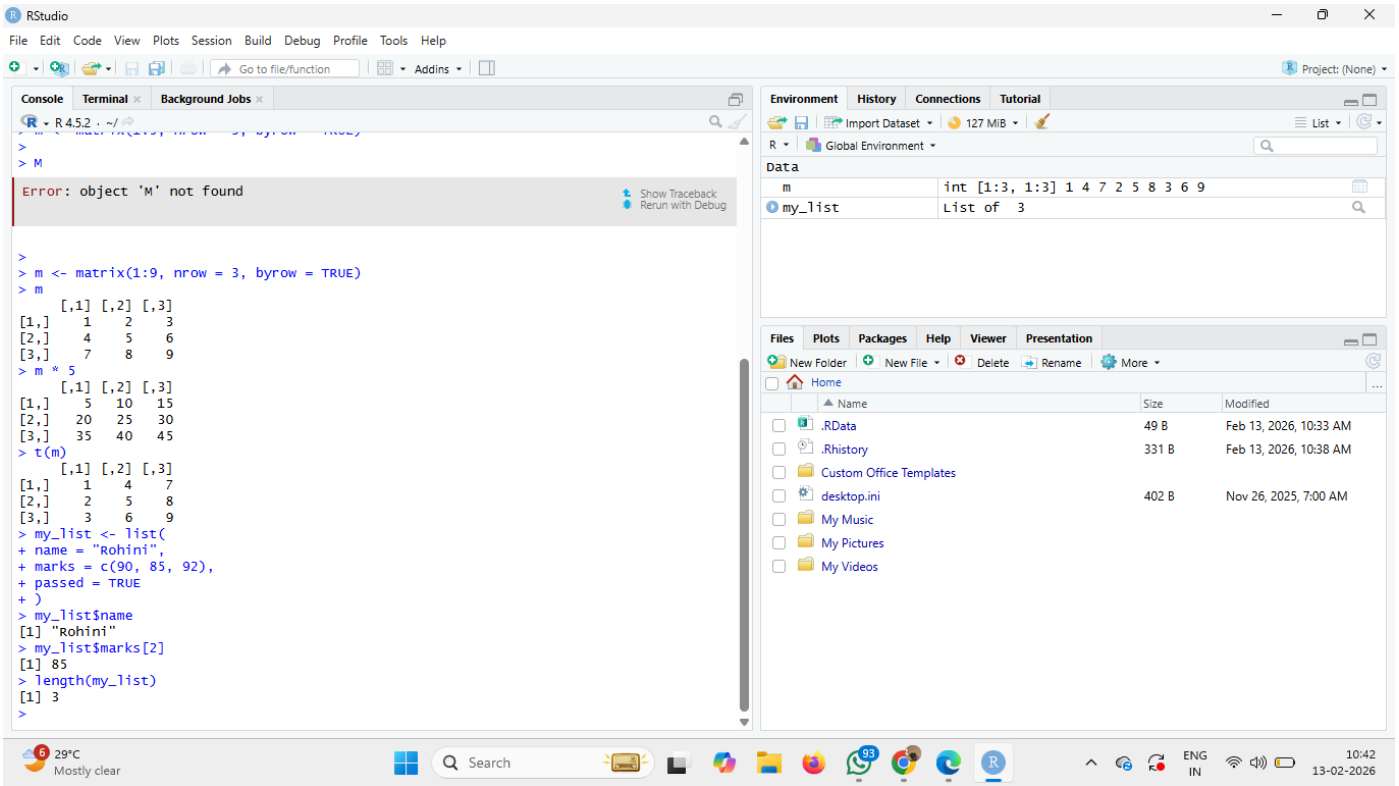
Step 4: Create a list `my_list` with name, marks, and passed status; access `my_list$name`, `my_list$marks[2]`, and find `length(my_list)`.

Step 5: Display the results and Stop the program.

SOURCE CODE:

```
m <- matrix(1:9, nrow = 3, byrow = TRUE)
m
m * 5
t(m) my_list <- list(
name = "Rohini",
marks = c(90, 85, 92),
passed = TRUE
)
my_list$name
my_list$marks[2]
length(my_list)
```

OUTPUT:



RESULT:

The above program is executed successfully .

EX. NO: 04 DATE:	USE 'DATA FRAMES' TO LOAD THE DATA FROM CSV FILES AND APPLY R FUNCTIONS FOR UNDERSTANDING THE STRUCTURE OF THE DATA IN DATA FRAMES

AIM:

To Use Data frame to load the data from csv files and apply R functions for understanding the structure of the data in data frames using R program .

ALGORITHM :

Step 1: Start the program.

Step 2: Load the CSV file using `df ← read.csv("data.csv")`.

Step 3: Check the structure of the data frame using `str(df)`.

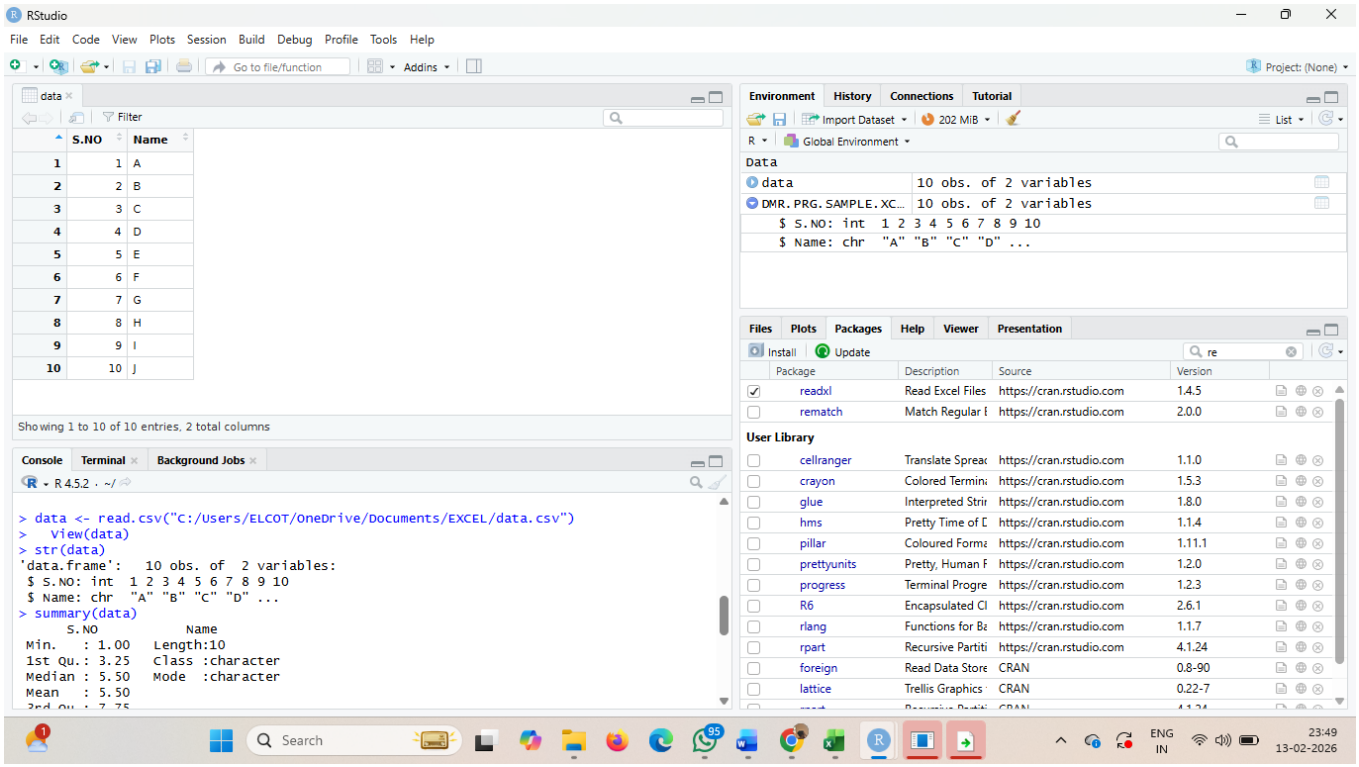
Step 4: Display summary and view data using `summary(df)`, `head(df)`, and `tail(df)`.

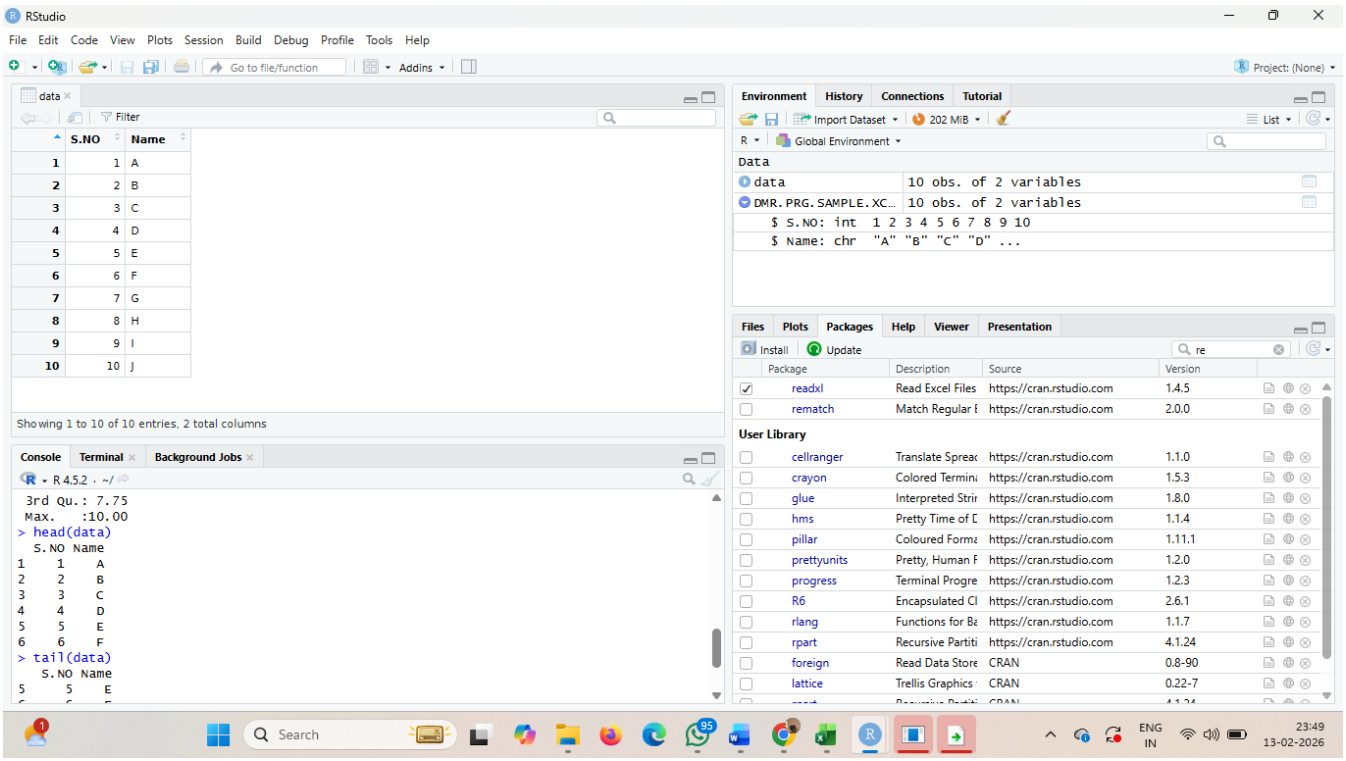
Step 5: Stop the program.

SOURCE CODE:

```
df <- read.csv("data.csv")
str(df)
summary(df)
head(df)
tail(df)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
data
Filter
S.NO Name
1 1 A
2 2 B
3 3 C
4 4 D
5 5 E
6 6 F
7 7 G
8 8 H
9 9 I
10 10 J
Showing 1 to 10 of 10 entries, 2 total columns
Environment History Connections Tutorial
Import Dataset 202 MiB
Global Environment
Data
data 10 obs. of 2 variables
DMR.PRG.SAMPLE.XC... 10 obs. of 2 variables
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
cellranger Translate Sprea https://cran.rstudio.com 1.1.0
crayon Colored Termin https://cran.rstudio.com 1.5.3
glue Interpreted Stri https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Console Terminal Background Jobs
R 4.5.2 · ~/
> data <- read.csv("C:/Users/ELCOT/OneDrive/Documents/EXCEL/data.csv")
> view(data)
> str(data)
'data.frame': 10 obs. of 2 variables:
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
> summary(data)
S.NO Name
Min. : 1.00 Length:10
1st Qu.: 3.25 Class :character
Median : 5.50 Mode :character
Mean : 5.50
Search
ENG IN
23:49
13-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
data
Filter
S.NO Name
1 1 A
2 2 B
3 3 C
4 4 D
5 5 E
6 6 F
7 7 G
8 8 H
9 9 I
10 10 J
Showing 1 to 10 of 10 entries, 2 total columns
Environment History Connections Tutorial
Import Dataset 202 MiB
Global Environment
Data
data 10 obs. of 2 variables
DMR.PRG.SAMPLE.XC... 10 obs. of 2 variables
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
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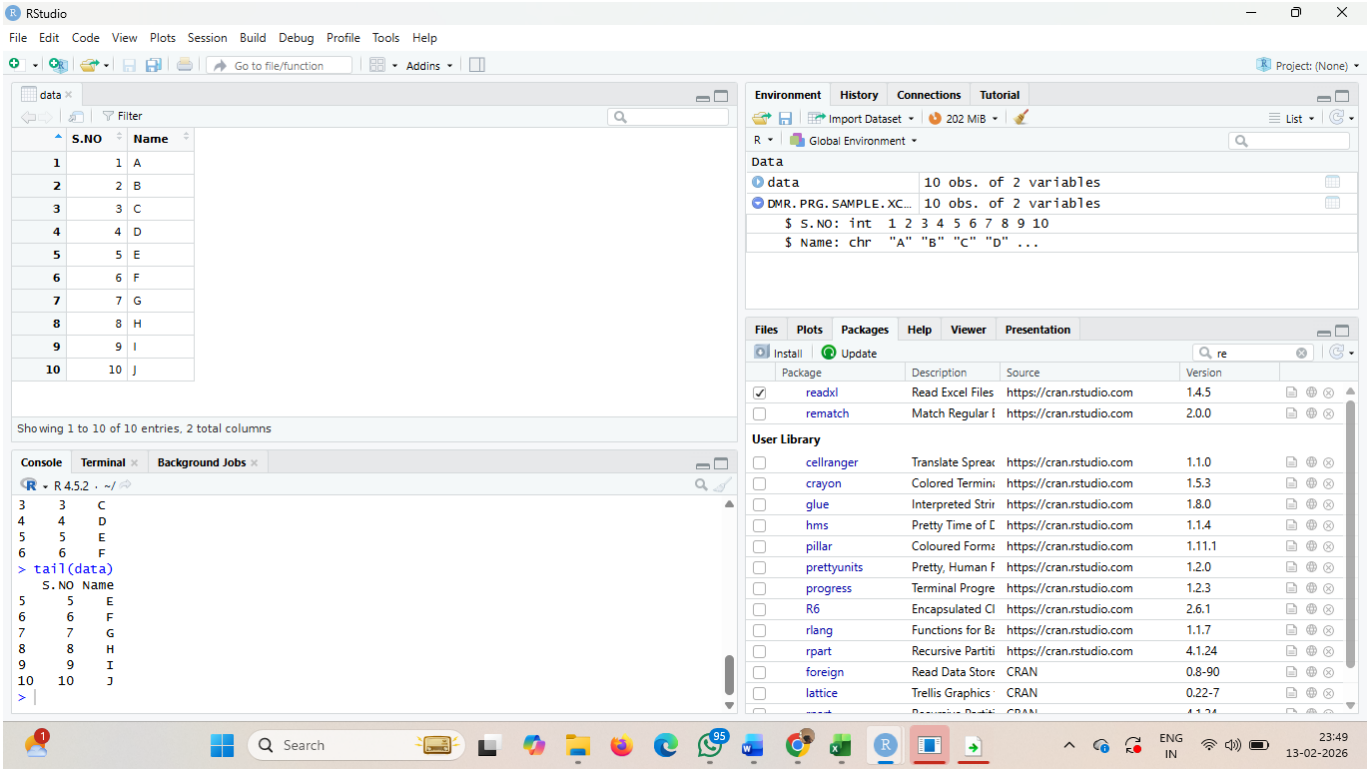
crayon Colored Termin https://cran.rstudio.com 1.5.3
glue Interpreted Stri https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Console Terminal Background Jobs
R 4.5.2 · ~/
3rd Qu.: 7.75
Max. :10.00
> head(data)
S.NO Name
1 1 A
2 2 B
3 3 C
4 4 D
5 5 E
6 6 F
> tail(data)
S.NO Name
5 5 E
Search
ENG IN
23:49
13-02-2026



RESULT:

The above program is executed successfully .

EX. NO: 05

DATE:

USE R FUNCTIONS TO OBTAIN DATA SUMMARY OF DATA FRAMES OBJECT

AIM:

To use R functions to obtain data summary of data frames object using R program .

ALGORITHM:

Step 1: Start the program.

Step 2: Load or have the data frame `df` ready.

Step 3: Display summary statistics using `summary(df)`.

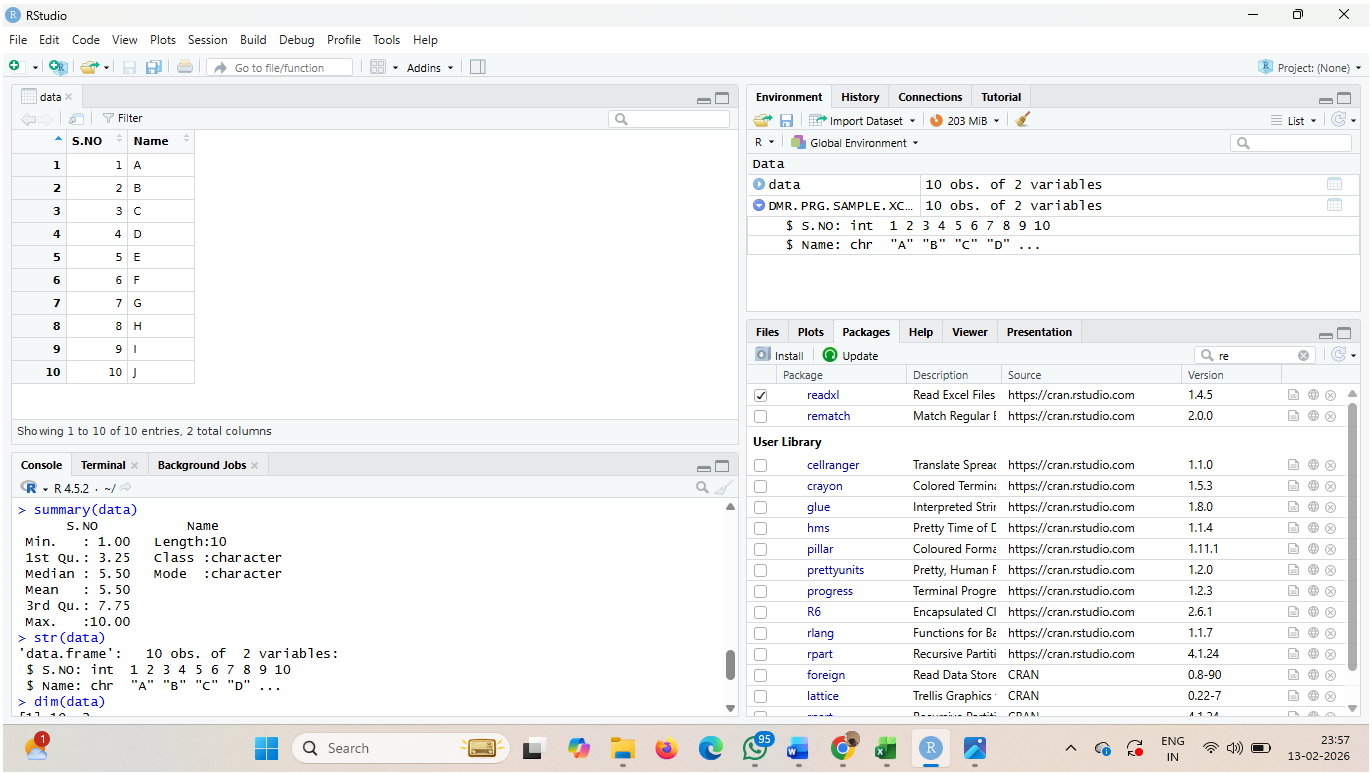
Step 4: Check structure and dimensions using `str(df)`, `dim(df)`, `nrow(df)`, `ncol(df)`.

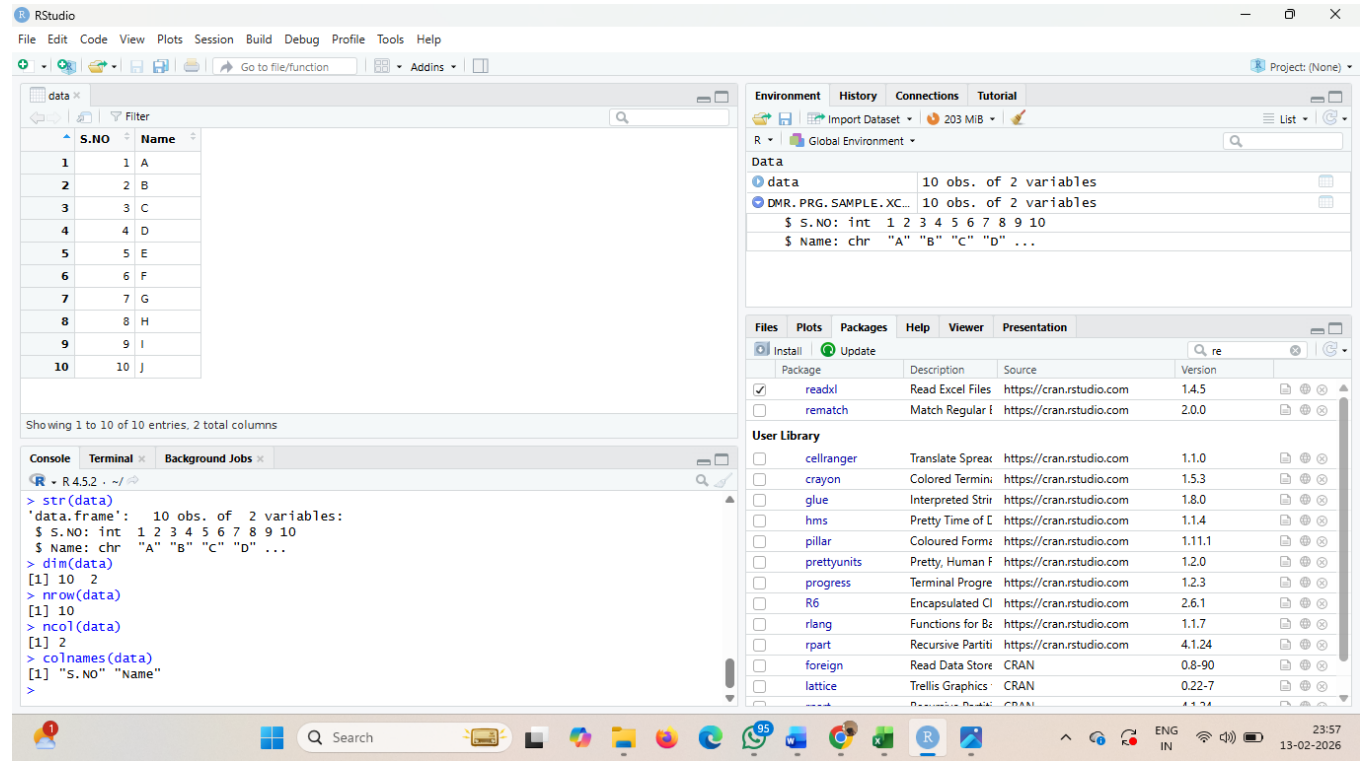
Step 5: Display column names using `colnames(df)` and Stop the program.

SOURCE CODE:

```
summary(df)
str(df)
dim(df)
nrow(df)
ncol(df)
colnames(df)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
data
S.NO Name
1 1 A
2 2 B
3 3 C
4 4 D
5 5 E
6 6 F
7 7 G
8 8 H
9 9 I
10 10 J
Showing 1 to 10 of 10 entries, 2 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
> summary(data)
S.NO Name
Min. : 1.00 Length:10
1st Qu.: 3.25 Class :character
Median : 5.50 Mode :character
Mean : 5.50
3rd Qu.: 7.75
Max. :10.00
> str(data)
'data.frame': 10 obs. of 2 variables:
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
> dim(data)
Environment History Connections Tutorial
Data
data 10 obs. of 2 variables
DMR.PRG.SAMPLE.XC... 10 obs. of 2 variables
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
crayon Colored Termina https://cran.rstudio.com 1.5.3
glue Interpreted Strin https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Search
ENG IN
23:57 13-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
data
S.NO Name
1 1 A
2 2 B
3 3 C
4 4 D
5 5 E
6 6 F
7 7 G
8 8 H
9 9 I
10 10 J
Showing 1 to 10 of 10 entries, 2 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
> str(data)
'data.frame': 10 obs. of 2 variables:
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
> dim(data)
[1] 10 2
> nrow(data)
[1] 10
> ncol(data)
[1] 2
> colnames(data)
[1] "S.NO" "Name"
>
Environment History Connections Tutorial
Data
data 10 obs. of 2 variables
DMR.PRG.SAMPLE.XC... 10 obs. of 2 variables
$ S.NO: int 1 2 3 4 5 6 7 8 9 10
$ Name: chr "A" "B" "C" "D" ...
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
crayon Colored Termina https://cran.rstudio.com 1.5.3
glue Interpreted Strin https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Search
ENG IN
23:57 13-02-2026

RESULT:

The above program is executed successfully.

EX. NO: 06 DATE:	USE A DATA SET AND DO STATISTICAL ANALYSIS ON THE DATA USING R

AIM:

To use a data set and do statistical analysis on the data using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load the dataset using `data("mtcars")`.

Step 3: Calculate measures of central tendency and dispersion:

- `mean(mtcars$mpg)`
- `median(mtcars$mpg)`
- `sd(mtcars$mpg)`
- `var(mtcars$mpg)`

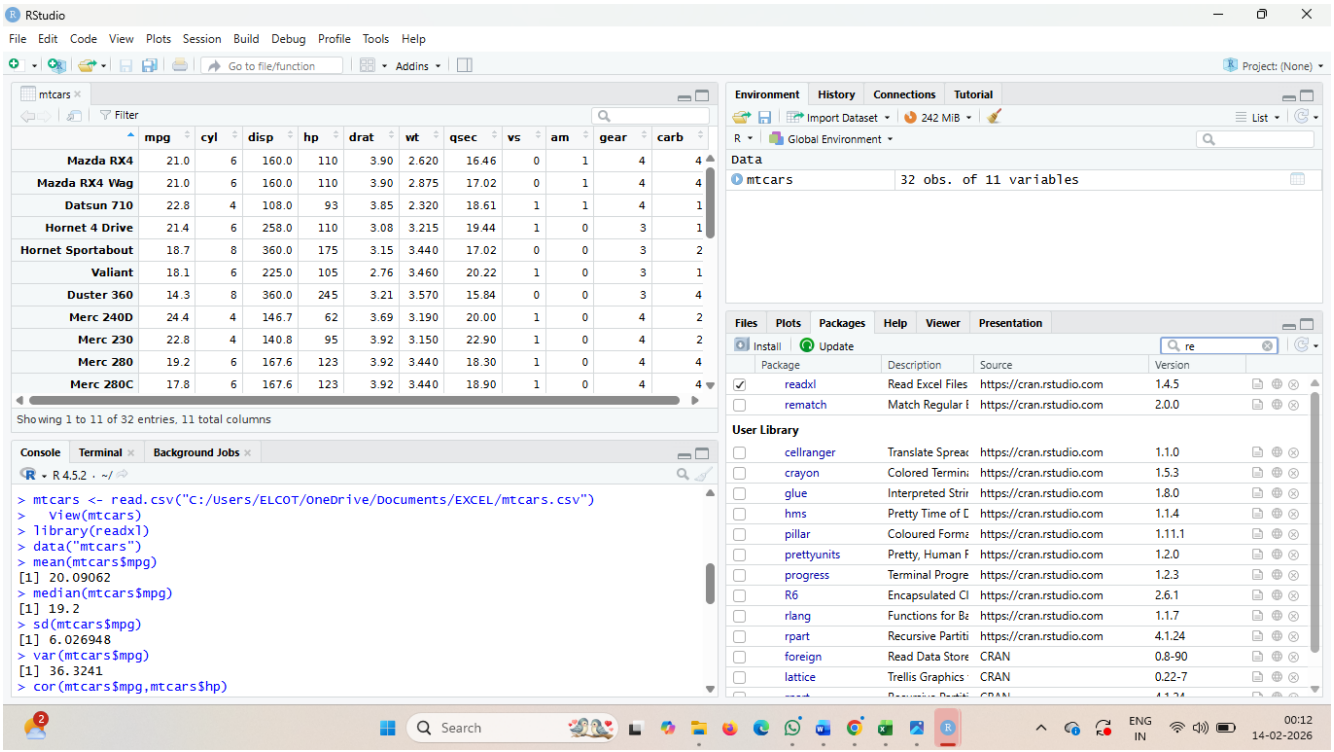
Step 4: Calculate correlation between variables: `cor(mtcars$mpg, mtcars$hp)`.

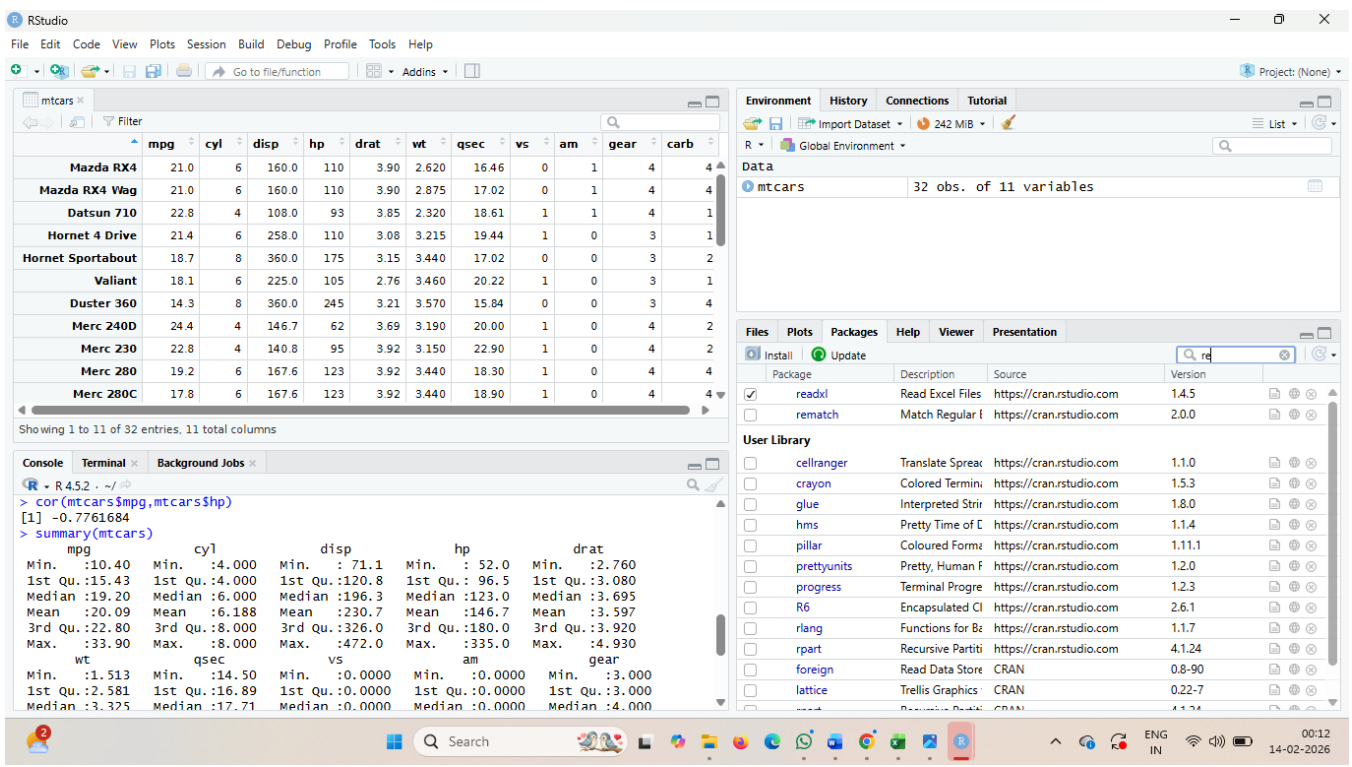
Step 5: Display the overall summary using `summary(mtcars)` and stop the program.

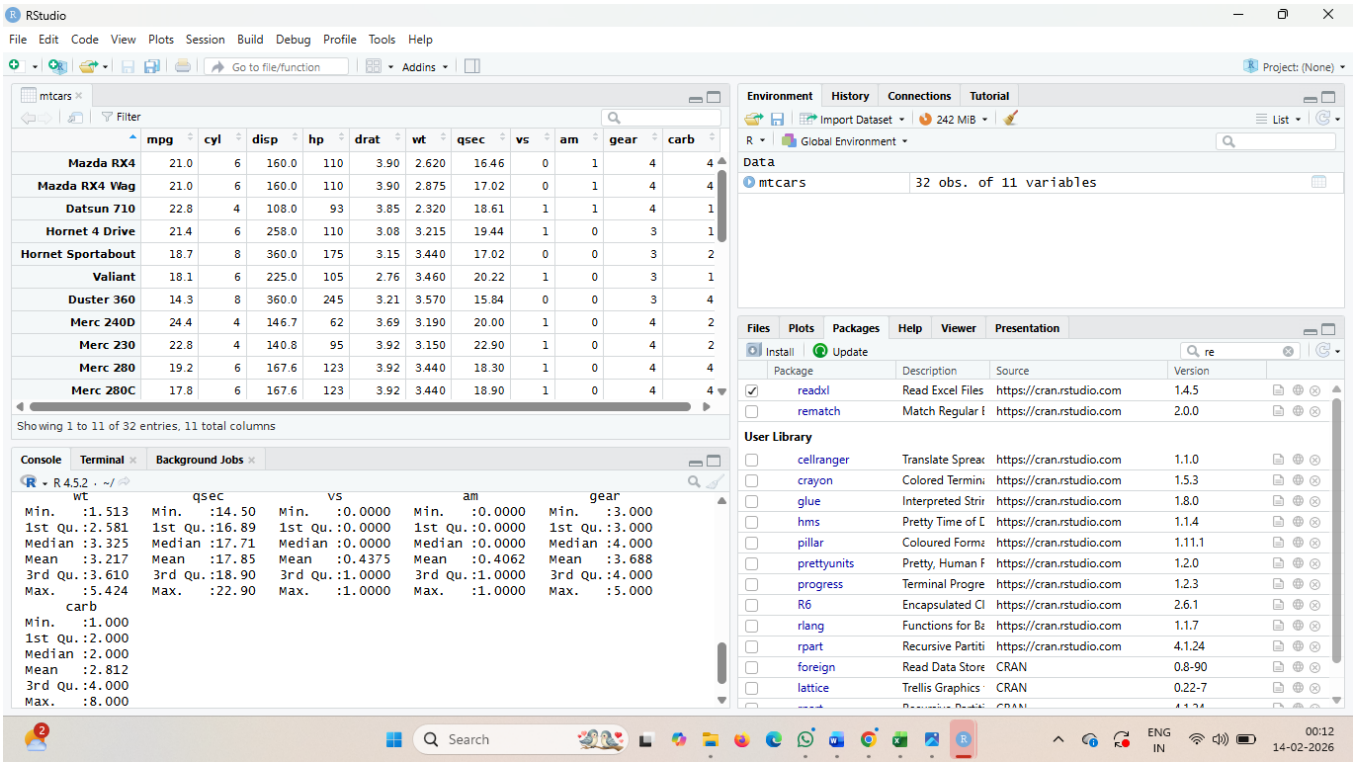
SOURCE CODE:

```
data("mtcars")
mean(mtcars$mpg)
median(mtcars$mpg)
sd(mtcars$mpg)
var(mtcars$mpg)
cor(mtcars$mpg, mtcars$hp)
summary(mtcars)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
mtcars
Filter
mpg cyl disp hp drat wt qsec vs am gear carb
Mazda RX4 21.0 6 160.0 110 3.90 2.620 16.46 0 1 4 4
Mazda RX4 Wag 21.0 6 160.0 110 3.90 2.875 17.02 0 1 4 4
Datsun 710 22.8 4 108.0 93 3.85 2.320 18.61 1 1 4 1
Hornet 4 Drive 21.4 6 258.0 110 3.08 3.215 19.44 1 0 3 1
Hornet Sportabout 18.7 8 360.0 175 3.15 3.440 17.02 0 0 3 2
Valiant 18.1 6 225.0 105 2.76 3.460 20.22 1 0 3 1
Duster 360 14.3 8 360.0 245 3.21 3.570 15.84 0 0 3 4
Merc 240D 24.4 4 146.7 62 3.69 3.190 20.00 1 0 4 2
Merc 230 22.8 4 140.8 95 3.92 3.150 22.90 1 0 4 2
Merc 280 19.2 6 167.6 123 3.92 3.440 18.30 1 0 4 4
Merc 280C 17.8 6 167.6 123 3.92 3.440 18.90 1 0 4 4
Showing 1 to 11 of 32 entries, 11 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
> mtcars <- read.csv("C:/Users/ELCOT/OneDrive/Documents/EXCEL/mtcars.csv")
> View(mtcars)
> library(readxl)
> data("mtcars")
> mean(mtcars$mpg)
[1] 20.09062
> median(mtcars$mpg)
[1] 19.2
> sd(mtcars$mpg)
[1] 6.026948
> var(mtcars$mpg)
[1] 36.3241
> cor(mtcars$mpg,mtcars$hp)
Environment History Connections Tutorial
Import Dataset 242 MiB
Global Environment
Data
mtcars 32 obs. of 11 variables
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
crayon Colored Termina https://cran.rstudio.com 1.5.3
glue Interpreted Strir https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Search
ENG IN
00:12 14-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
mtcars
Filter
mpg cyl disp hp drat wt qsec vs am gear carb
Mazda RX4 21.0 6 160.0 110 3.90 2.620 16.46 0 1 4 4
Mazda RX4 Wag 21.0 6 160.0 110 3.90 2.875 17.02 0 1 4 4
Datsun 710 22.8 4 108.0 93 3.85 2.320 18.61 1 1 4 1
Hornet 4 Drive 21.4 6 258.0 110 3.08 3.215 19.44 1 0 3 1
Hornet Sportabout 18.7 8 360.0 175 3.15 3.440 17.02 0 0 3 2
Valiant 18.1 6 225.0 105 2.76 3.460 20.22 1 0 3 1
Duster 360 14.3 8 360.0 245 3.21 3.570 15.84 0 0 3 4
Merc 240D 24.4 4 146.7 62 3.69 3.190 20.00 1 0 4 2
Merc 230 22.8 4 140.8 95 3.92 3.150 22.90 1 0 4 2
Merc 280 19.2 6 167.6 123 3.92 3.440 18.30 1 0 4 4
Merc 280C 17.8 6 167.6 123 3.92 3.440 18.90 1 0 4 4
Showing 1 to 11 of 32 entries, 11 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
> cor(mtcars$mpg,mtcars$hp)
[1] -0.7761684
> summary(mtcars)
mpg cyl disp hp drat
Min. :10.40 Min. :4.000 Min. : 71.1 Min. : 52.0 Min. :2.760
1st Qu.:15.43 1st Qu.:4.000 1st Qu.:120.8 1st Qu.: 96.5 1st Qu.:3.080
Median :19.20 Median :6.000 Median :196.3 Median :123.0 Median :3.695
Mean :20.09 Mean :6.188 Mean :230.7 Mean :146.7 Mean :3.597
3rd Qu.:22.80 3rd Qu.:8.000 3rd Qu.:326.0 3rd Qu.:180.0 3rd Qu.:3.920
Max. :33.90 Max. :8.000 Max. :472.0 Max. :335.0 Max. :4.930
wt qsec vs am gear
Min. :1.513 Min. :14.50 Min. :0.0000 Min. :0.0000 Min. :3.000
1st Qu.:2.581 1st Qu.:16.89 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:3.000
Median :3.325 Median :17.71 Median :0.0000 Median :0.0000 Median :4.000
Environment History Connections Tutorial
Import Dataset 242 MiB
Global Environment
Data
mtcars 32 obs. of 11 variables
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
readxl Read Excel Files https://cran.rstudio.com 1.4.5
rematch Match Regular E https://cran.rstudio.com 2.0.0
User Library
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
crayon Colored Termina https://cran.rstudio.com 1.5.3
glue Interpreted Strir https://cran.rstudio.com 1.8.0
hms Pretty Time of D https://cran.rstudio.com 1.1.4
pillar Coloured Forma https://cran.rstudio.com 1.11.1
prettyunits Pretty, Human F https://cran.rstudio.com 1.2.0
progress Terminal Progre https://cran.rstudio.com 1.2.3
R6 Encapsulated Cl https://cran.rstudio.com 2.6.1
rlang Functions for Ba https://cran.rstudio.com 1.1.7
rpart Recursive Partiti https://cran.rstudio.com 4.1.24
foreign Read Data Store CRAN 0.8-90
lattice Trellis Graphics CRAN 0.22-7
Search
ENG IN
00:12 14-02-2026



RESULT:

The above program is executed successfully.

EX. NO: 07 DATE:	APPLY R FUNCTIONS TO GENERATE HISTOGRAM, BAR PLOT AND DENSITY PLOT GRAPHS

AIM:

To apply functions to generate histogram bar plot and density plot graphs using R program .

ALGORITHM:

Step 1: Start the program.

Step 2: Load the dataset using `data("mtcars")`.

Step 3: Create a histogram of `mpg` using `hist(mtcars$mpg)`.

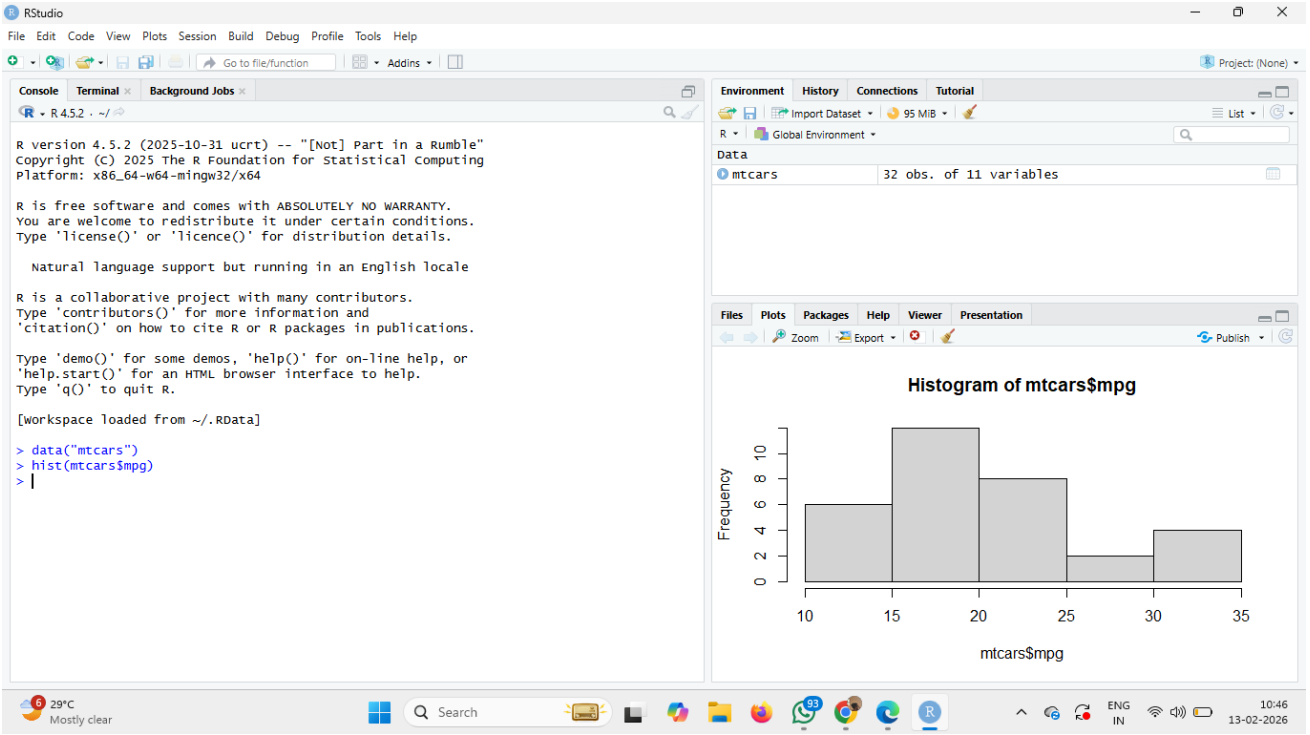
Step 4: Create a bar plot of `hp` using `barplot(mtcars$hp)`.

Step 5: Create a density plot of `mpg` using `plot(density(mtcars$mpg))` and Stop the program.

SOURCE CODE:

```
data("mtcars")
hist(mtcars$mpg)
barplot(mtcars$hp)
plot(density(mtcars$mpg))
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Rumble"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
R is free software and comes with ABSOLUTELY NO WARRANTY.
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Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
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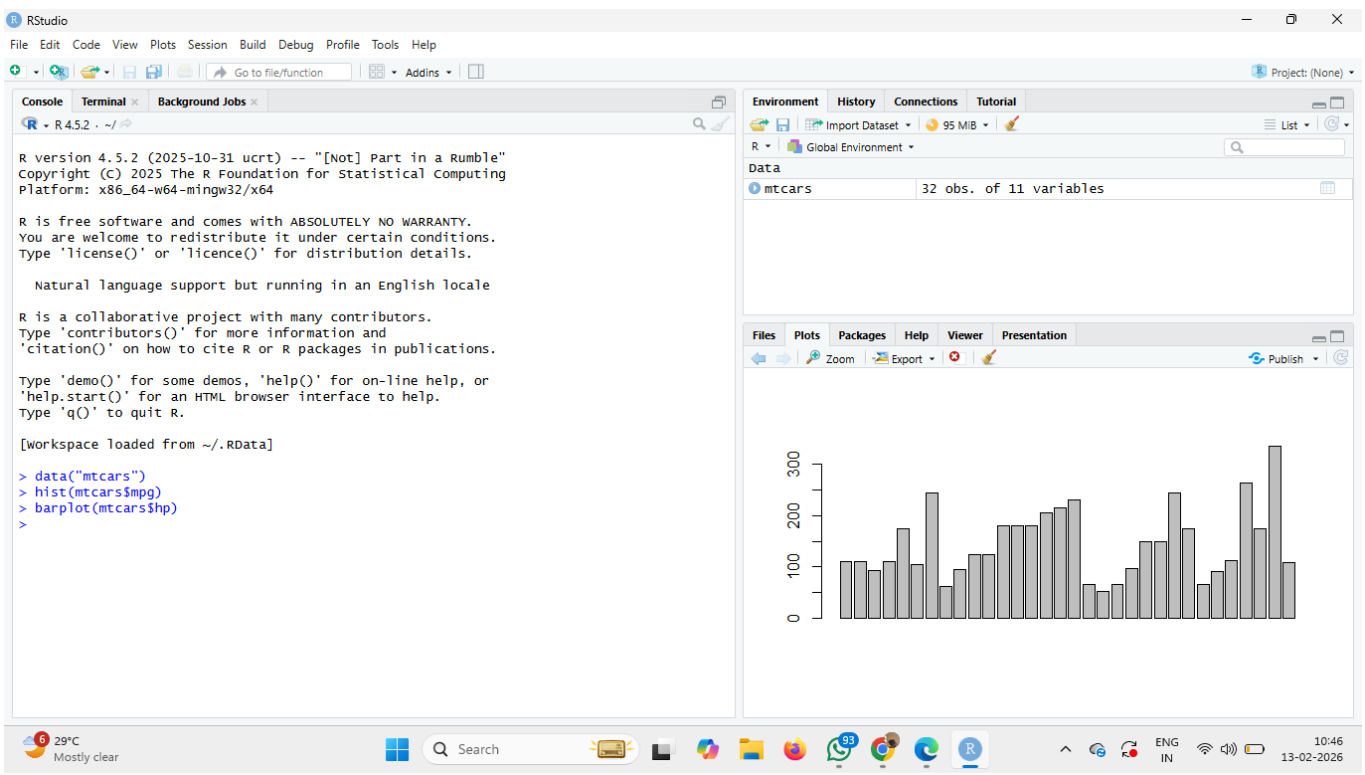
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> data("mtcars")
> hist(mtcars$mpg)
>
Environment History Connections Tutorial
Import Dataset 95 MiB
Global Environment
Data
mtcars 32 obs. of 11 variables
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Histogram of mtcars$mpg
Frequency
0 2 4 6 8 10
10 15 20 25 30 35
mtcars$mpg
29°C Mostly clear
Search
ENG IN
10:46 13-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Rumble"
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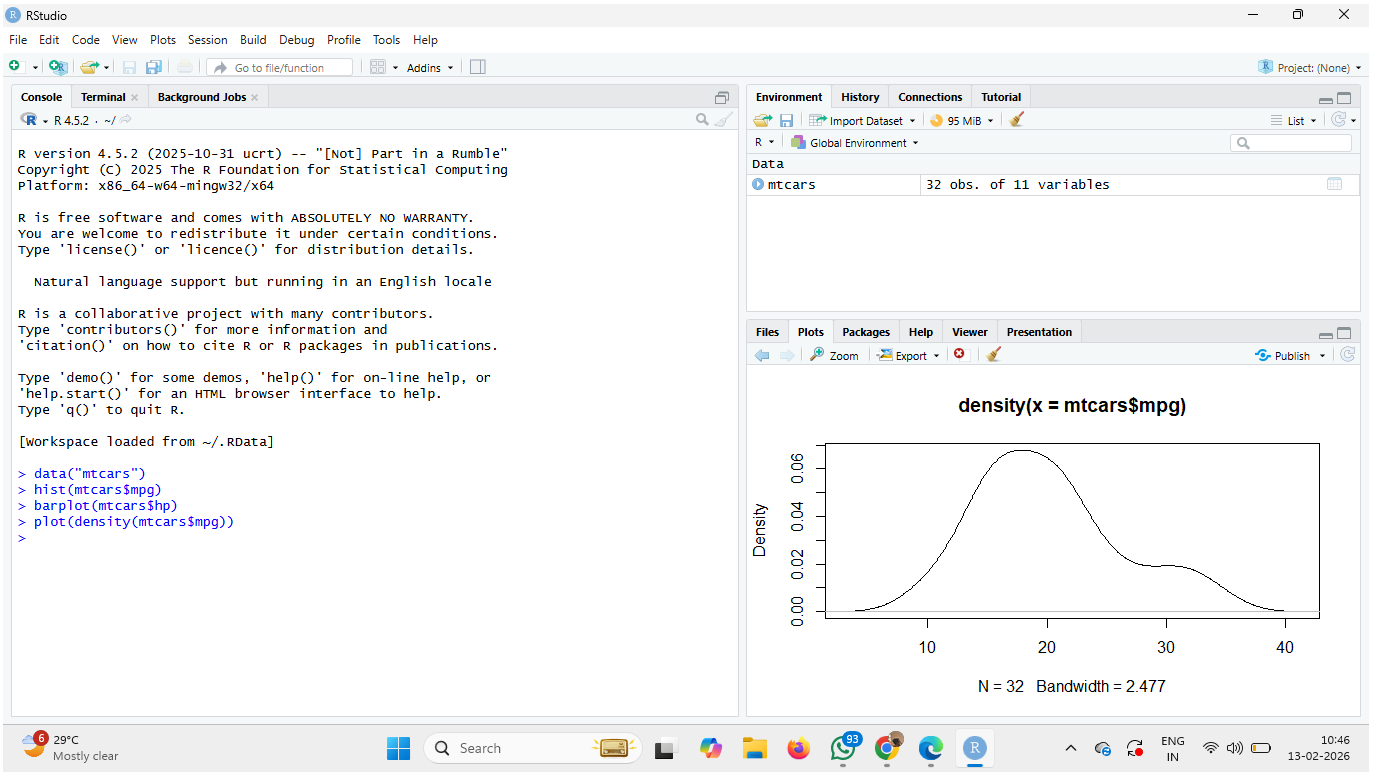
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> data("mtcars")
> hist(mtcars$mpg)
> barplot(mtcars$hp)
>
Environment History Connections Tutorial
Import Dataset 95 MiB
Global Environment
Data
mtcars 32 obs. of 11 variables
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
0 100 200 300
29°C Mostly clear
Search
ENG IN
10:46 13-02-2026



RESULT:

The above program is executed successfully.

EX. NO: 08 DATE:	**DEMONSTRATE THE LINEAR RELATIONSHIP MODEL BETWEEN THE DISTANCE AND SPEED OF THE 'CARS' DATASET.AND VISUALIZE THE LINEAR REGRESSION GRAPHICALLY BY PLOTTING A CHART**

AIM:

To Demonstrate the linear relationship model between the distance and speed of the "cars" dataset. And visualize Linear Regression graphically by plotting a chart using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load the dataset using `data("cars")`.

Step 3: Create a linear regression model: `model ← lm(dist ~ speed, data = cars)`.

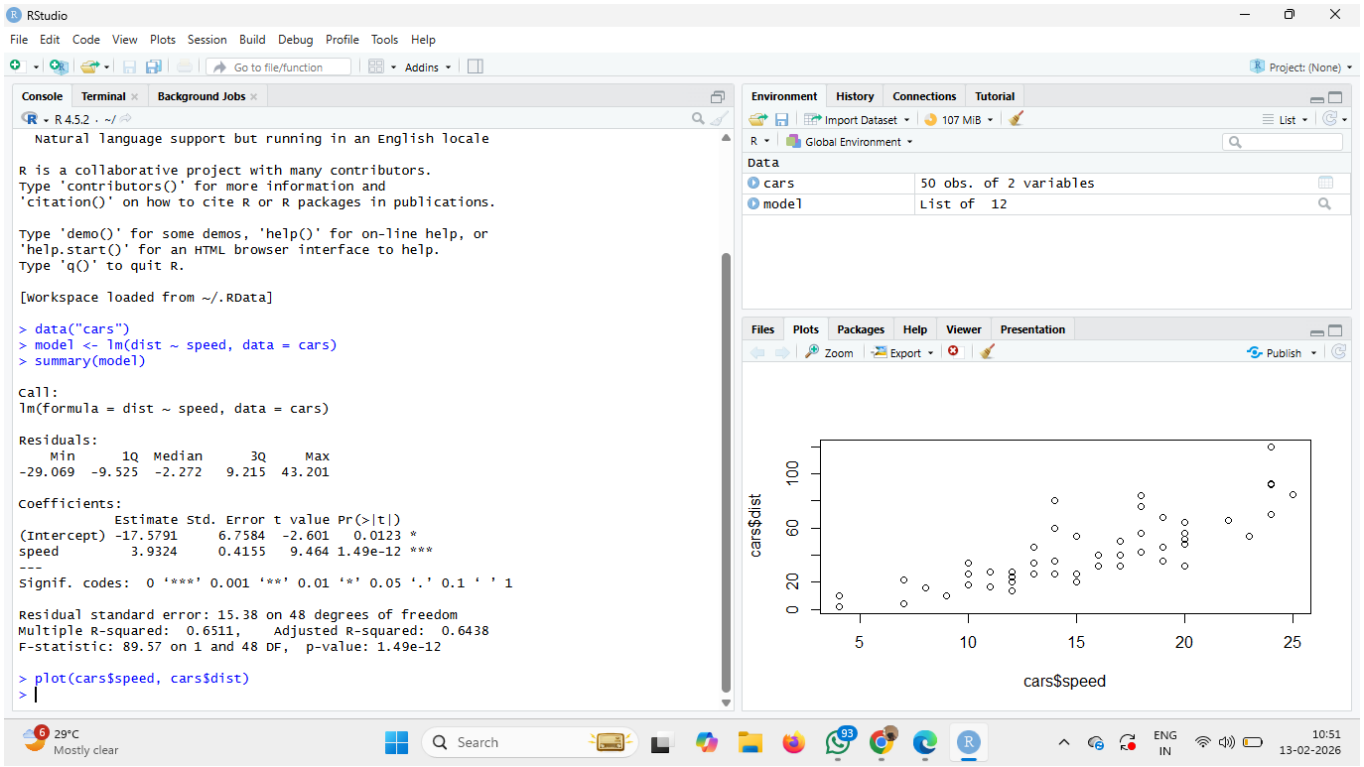
Step 4: View the model summary using `a summary(model)`.

Step 5: Plot the data and regression line using `plot (cars$speed, cars$dist)` and `abline(model)`, then stop the program.

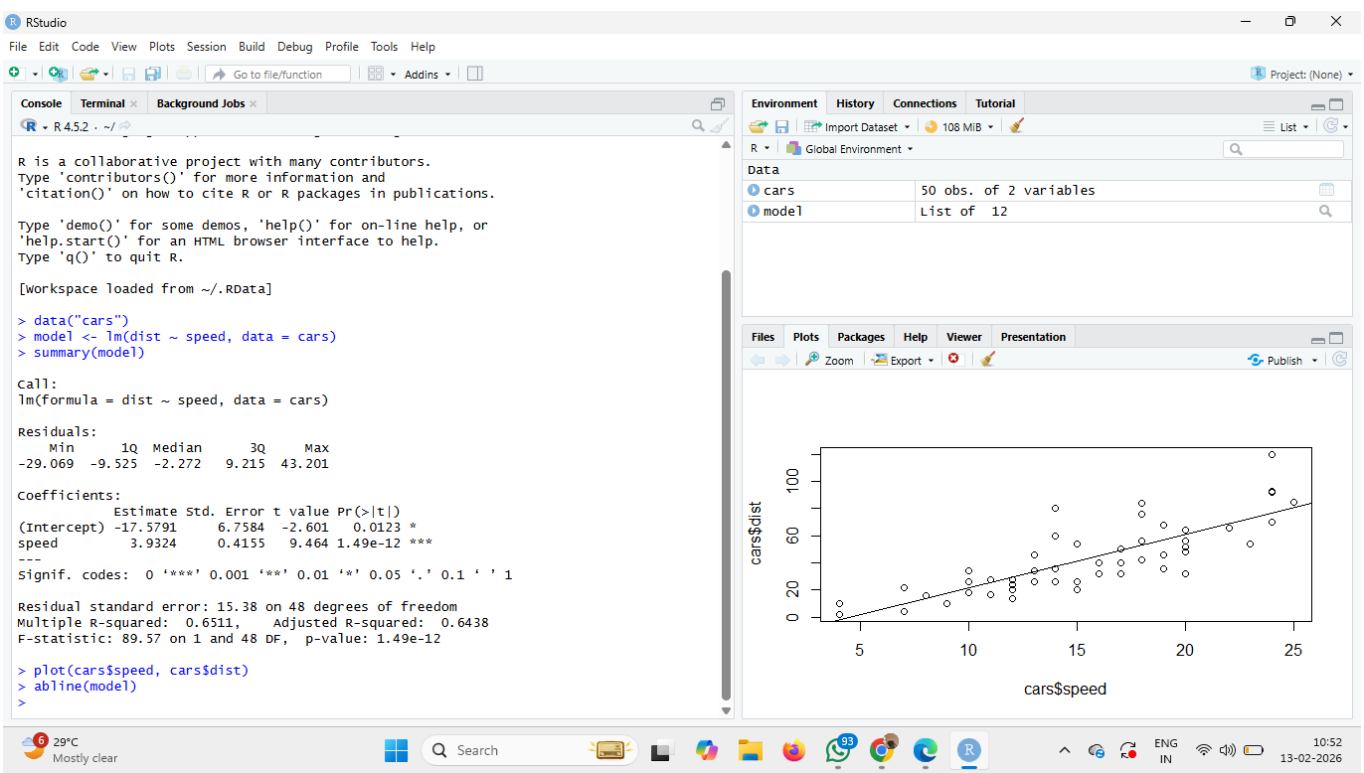
SOURCE CODE:

```
data("cars")
model <- lm(dist ~ speed, data = cars)
summary(model)
plot (cars$speed, cars$dist)
abline(model)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> data("cars")
> model <- lm(dist ~ speed, data = cars)
> summary(model)
Call:
lm(formula = dist ~ speed, data = cars)
Residuals:
Min 1Q Median 3Q Max
-29.069 -9.525 -2.272 9.215 43.201
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791 6.7584 -2.601 0.0123 *
speed 3.9324 0.4155 9.464 1.49e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared: 0.6511, Adjusted R-squared: 0.6438
F-statistic: 89.57 on 1 and 48 DF, p-value: 1.49e-12
> plot(cars$speed, cars$dist)
Environment History Connections Tutorial
Import Dataset 107 MiB
Global Environment
Data
cars 50 obs. of 2 variables
model List of 12
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
cars$dist
cars$speed
0 20 60 100
5 10 15 20 25
29°C Mostly clear
Search
ENG IN
10:51 13-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
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'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> data("cars")
> model <- lm(dist ~ speed, data = cars)
> summary(model)
Call:
lm(formula = dist ~ speed, data = cars)
Residuals:
Min 1Q Median 3Q Max
-29.069 -9.525 -2.272 9.215 43.201
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791 6.7584 -2.601 0.0123 *
speed 3.9324 0.4155 9.464 1.49e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared: 0.6511, Adjusted R-squared: 0.6438
F-statistic: 89.57 on 1 and 48 DF, p-value: 1.49e-12
> plot(cars$speed, cars$dist)
> abline(model)
Environment History Connections Tutorial
Import Dataset 108 MiB
Global Environment
Data
cars 50 obs. of 2 variables
model List of 12
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
cars$dist
cars$speed
0 20 60 100
5 10 15 20 25
29°C Mostly clear
Search
ENG IN
10:52 13-02-2026

RESULT:

The above program is executed successfully.

EX. NO: 09

DATE:

DEMONSTRATE DECISION TREE ALGORITHM USING "STUDENT" DATASET

AIM:

To Demonstrate Decision Tree algorithm using "student" dataset using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load required libraries: `library(rpart)` and `library(rpart.plot)`.

Step 3: Load the dataset using `student ← read.csv("student.csv")`.

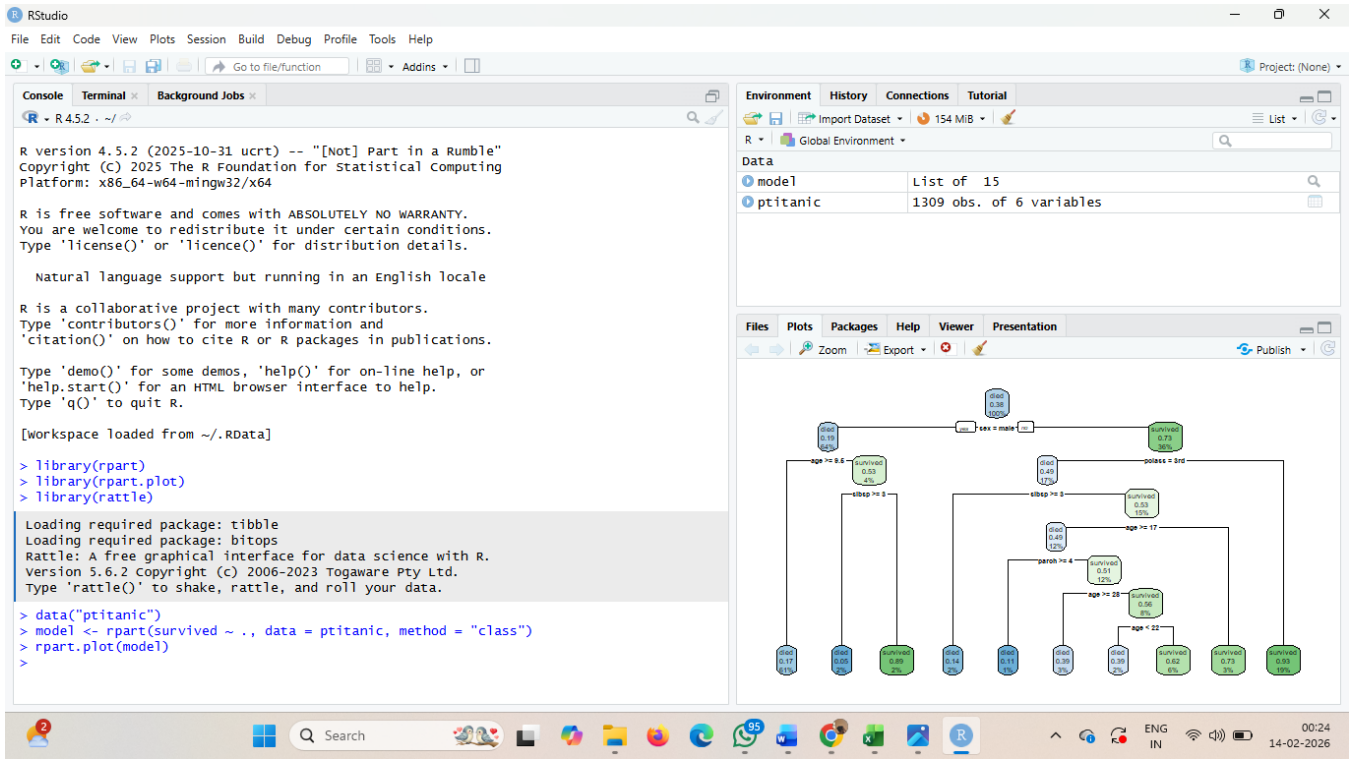
Step 4: Create a decision tree model: `model ← rpart(Result ~ ., data = student, method="class")`.

Step 5: Visualize the decision tree using `rpart.plot(model)` and Stop the program.

SOURCE CODE:

```
library(rpart)
library(rpart.plot)
data(“ptitanic”)
model <- rpart(Survived ~., data = ptitanic, method="class")
rpart.plot(model)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Rumble"
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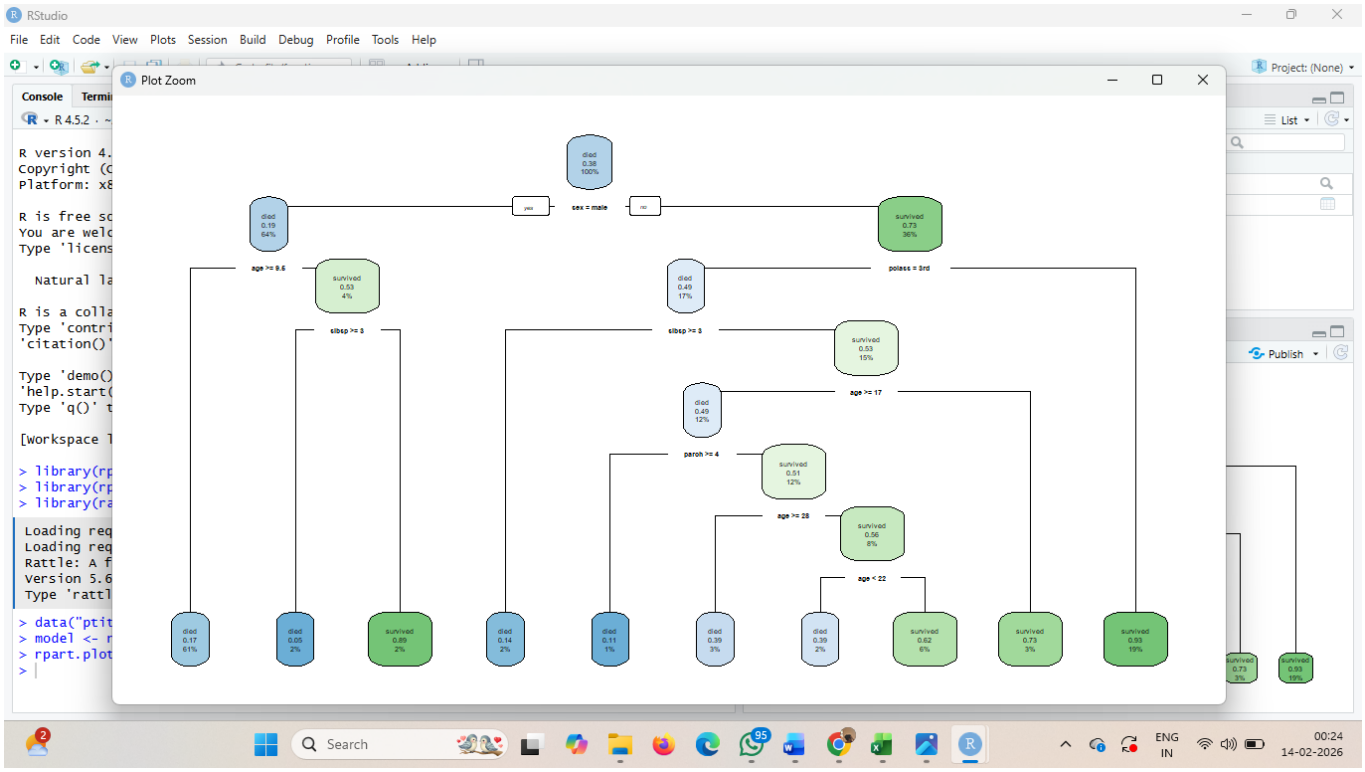
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> library(rpart)
> library(rpart.plot)
> library(rattle)
Loading required package: tibble
Loading required package: bitops
Rattle: A free graphical interface for data science with R.
Version 5.6.2 Copyright (c) 2006-2023 Togaware Pty Ltd.
Type 'rattle()' to shake, rattle, and roll your data.
> data("ptitanic")
> model <- rpart(survived ~ ., data = ptitanic, method = "class")
> rpart.plot(model)
>
Environment History Connections Tutorial
Import Dataset 154 MiB
Global Environment
Data
model List of 15
ptitanic 1309 obs. of 6 variables
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Search
ENG IN
00:24
14-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Plot Zoom
Console
Project: (None)
Publish
Search
ENG IN
00:24
14-02-2026

RESULT:

The above program is executed successfully.

EX. NO: 10

DATE:

DEMONSTRATE K-MEANS ALGORITHM USING "IRIS" DATASET

AIM:

To Demonstrate K-Means algorithm using "iris" dataset using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load the dataset using `data("iris")`.

Step 3: Perform K-Means clustering on the first four columns: `km ← kmeans(iris[,1:4], centers=3)`.

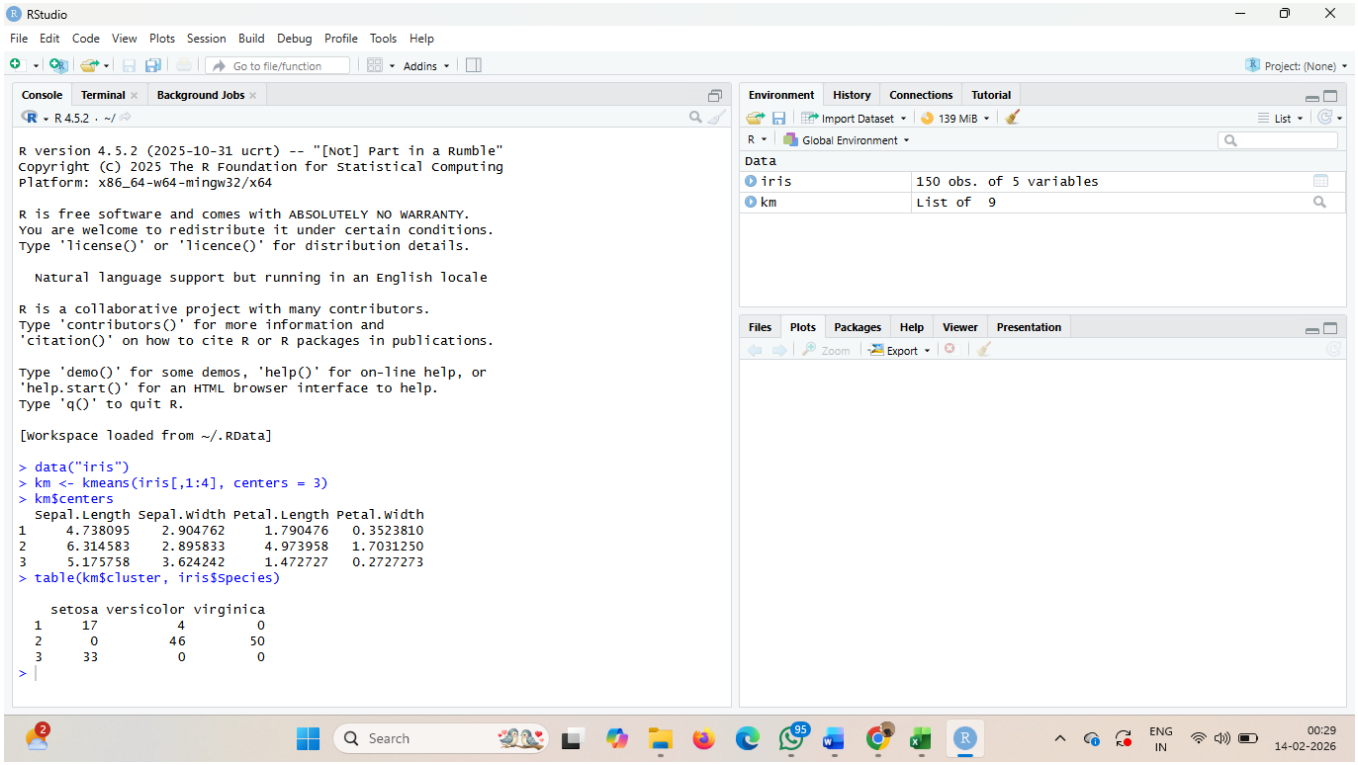
Step 4: View the cluster centers using `km$centers`.

Step 5: Compare clusters with actual species using `table(km$cluster, iris$Species)` and Stop the program.

SOURCE CODE:

```
data("iris")
km <- kmeans(iris [,1:4], centers=3)
km$centers
table (km$cluster, iris$Species)
```

OUTPUT:



RESULT:

The above program is executed successfully.

EX. NO: 11 DATE:	DEMONSTRATE SVM USING “EMPLOYEE” DATASET

AIM:

To Demonstrate SVM using "employee" dataset using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load required `library(e1071)`.

Step 3: Load the dataset using `employee ← read.csv("employee.csv")`.

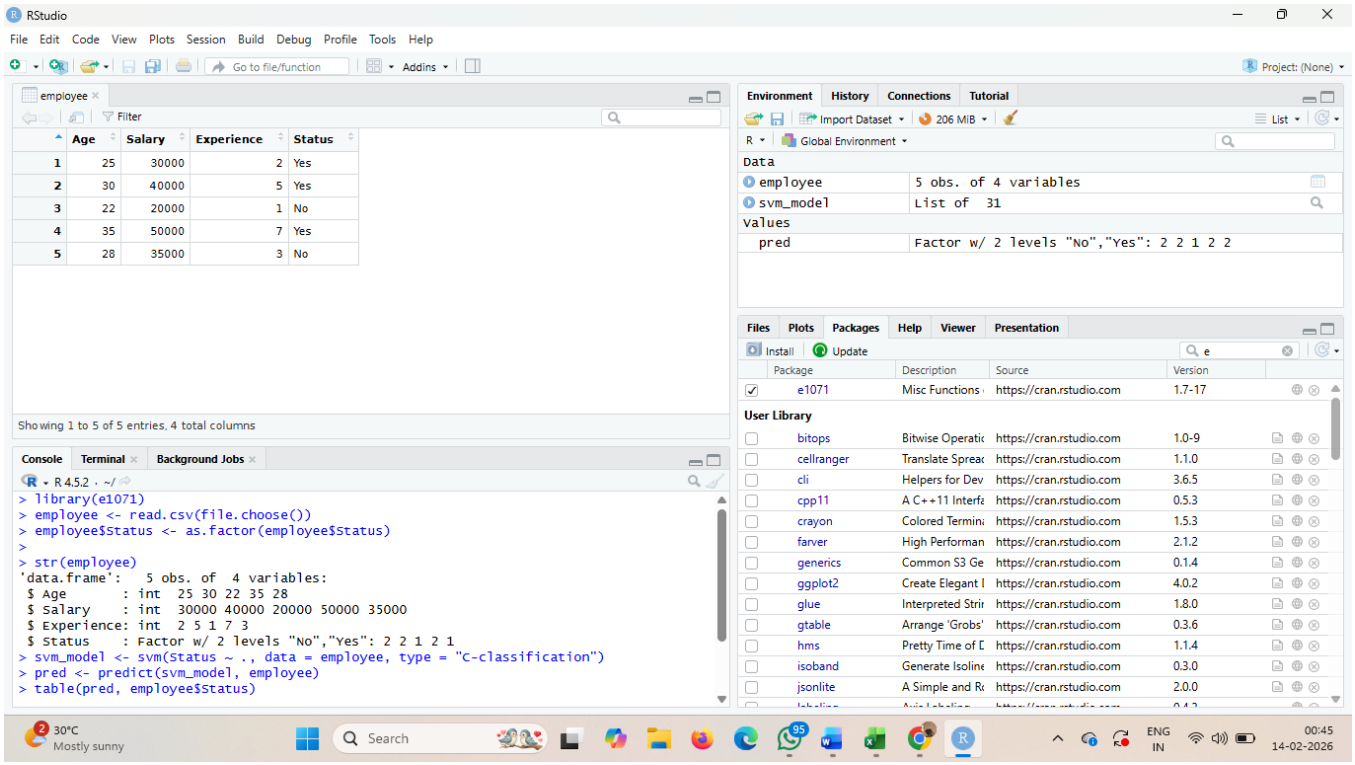
Step 4: Create the SVM model: `svm_model ← svm(Status ~., data=employee)` and make predictions: `pred ← predict (svm_model, employee)`.

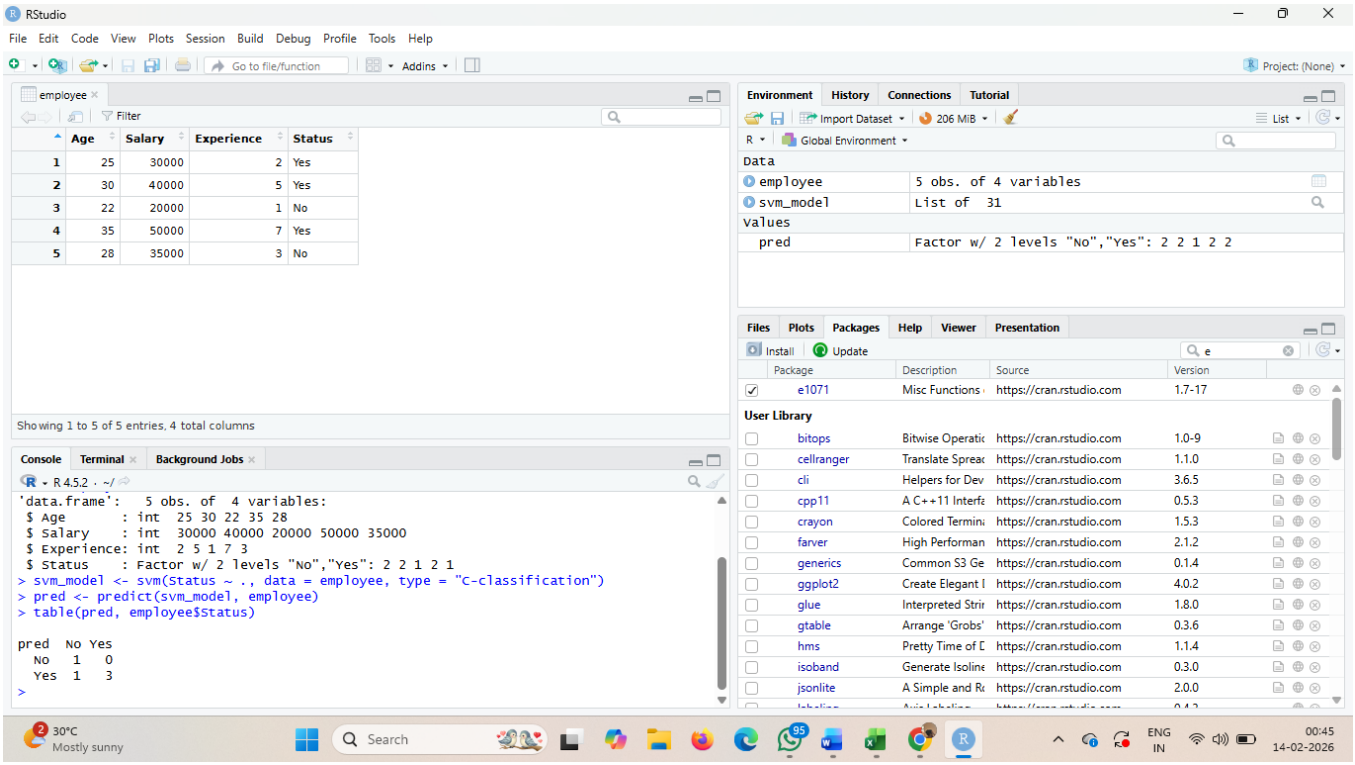
Step 5: Compare predictions with actual status using `table (pred, employee$Status)` and stop the program.

SOURCE CODE:

```
library(e1071)
employee <- read.csv("employee.csv")
employee$Status <- as.factor(employee$Status)
str(employee)
svm_model <- svm(Status ~., data = employee, type = "C-classification")
pred <- predict (svm_model, employee)
table (pred, employee$Status)
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
employee
Age Salary Experience Status
1 25 30000 2 Yes
2 30 40000 5 Yes
3 22 20000 1 No
4 35 50000 7 Yes
5 28 35000 3 No
Showing 1 to 5 of 5 entries, 4 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
> library(e1071)
> employee <- read.csv(file.choose())
> employee$Status <- as.factor(employee$Status)
>
> str(employee)
'data.frame': 5 obs. of 4 variables:
$ Age : int 25 30 22 35 28
$ Salary : int 30000 40000 20000 50000 35000
$ Experience: int 2 5 1 7 3
$ Status : Factor w/ 2 levels "No","Yes": 2 2 1 2 1
> svm_model <- svm(Status ~ ., data = employee, type = "C-classification")
> pred <- predict(svm_model, employee)
> table(pred, employee$Status)
Environment History Connections Tutorial
Data
employee 5 obs. of 4 variables
svm_model List of 31
values
pred Factor w/ 2 levels "No","Yes": 2 2 1 2 2
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
e1071 Misc Functions https://cran.rstudio.com 1.7-17
User Library
bitops Bitwise Operatic https://cran.rstudio.com 1.0-9
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
cli Helpers for Dev https://cran.rstudio.com 3.6.5
cpp11 A C++11 Interfa https://cran.rstudio.com 0.5.3
crayon Colored Termina https://cran.rstudio.com 1.5.3
farver High Performan https://cran.rstudio.com 2.1.2
generics Common S3 Ge https://cran.rstudio.com 0.1.4
ggplot2 Create Elegant I https://cran.rstudio.com 4.0.2
glue Interpreted Strir https://cran.rstudio.com 1.8.0
gtable Arrange 'Grobs' https://cran.rstudio.com 0.3.6
hms Pretty Time of D https://cran.rstudio.com 1.1.4
isoband Generate Isoline https://cran.rstudio.com 0.3.0
jsonlite A Simple and Ro https://cran.rstudio.com 2.0.0
30°C Mostly sunny
Search
ENG IN
00:45 14-02-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
employee
Age Salary Experience Status
1 25 30000 2 Yes
2 30 40000 5 Yes
3 22 20000 1 No
4 35 50000 7 Yes
5 28 35000 3 No
Showing 1 to 5 of 5 entries, 4 total columns
Console Terminal Background Jobs
R 4.5.2 · ~/
'data.frame': 5 obs. of 4 variables:
$ Age : int 25 30 22 35 28
$ Salary : int 30000 40000 20000 50000 35000
$ Experience: int 2 5 1 7 3
$ Status : Factor w/ 2 levels "No","Yes": 2 2 1 2 1
> svm_model <- svm(Status ~ ., data = employee, type = "C-classification")
> pred <- predict(svm_model, employee)
> table(pred, employee$Status)
pred No Yes
No 1 0
Yes 1 3
>
Environment History Connections Tutorial
Data
employee 5 obs. of 4 variables
svm_model List of 31
values
pred Factor w/ 2 levels "No","Yes": 2 2 1 2 2
Files Plots Packages Help Viewer Presentation
Install Update
Package Description Source Version
e1071 Misc Functions https://cran.rstudio.com 1.7-17
User Library
bitops Bitwise Operatic https://cran.rstudio.com 1.0-9
cellranger Translate Spreac https://cran.rstudio.com 1.1.0
cli Helpers for Dev https://cran.rstudio.com 3.6.5
cpp11 A C++11 Interfa https://cran.rstudio.com 0.5.3
crayon Colored Termina https://cran.rstudio.com 1.5.3
farver High Performan https://cran.rstudio.com 2.1.2
generics Common S3 Ge https://cran.rstudio.com 0.1.4
ggplot2 Create Elegant I https://cran.rstudio.com 4.0.2
glue Interpreted Strir https://cran.rstudio.com 1.8.0
gtable Arrange 'Grobs' https://cran.rstudio.com 0.3.6
hms Pretty Time of D https://cran.rstudio.com 1.1.4
isoband Generate Isoline https://cran.rstudio.com 0.3.0
jsonlite A Simple and Ro https://cran.rstudio.com 2.0.0
30°C Mostly sunny
Search
ENG IN
00:45 14-02-2026

RESULT:

The above program is executed successfully.

EX. NO: 12

DATE:

IMPLEMENT APRIORI ALGORITHM USING "WHOLESALER" DATASET

AIM:

To Implement Apriori algorithm using "wholesaler" dataset using R program.

ALGORITHM:

Step 1: Start the program.

Step 2: Load the required library: `library(arules)`.

Step 3: Load the dataset as transactions: `data ← read.transactions("wholesaler.csv", format="basket", sep=",")`.

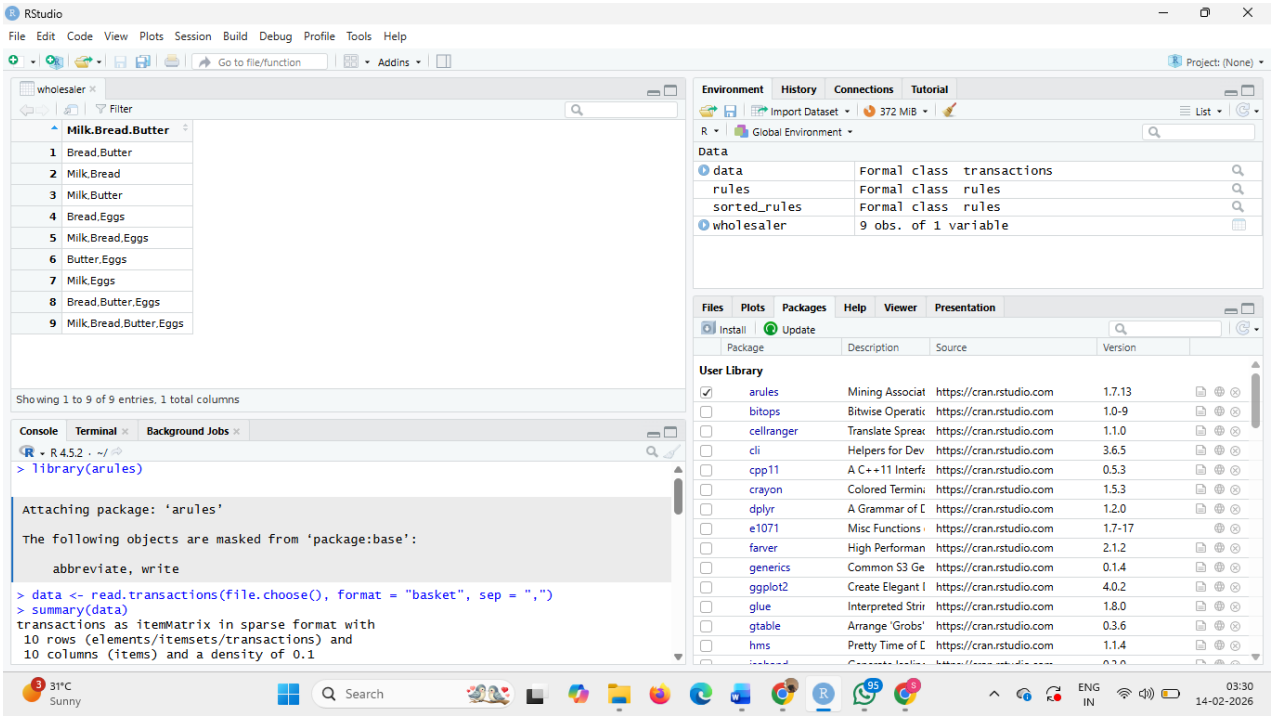
Step 4: Generate association rules using Apriori: `rules ← apriori(data, parameter=list (supp=0.001, conf=0.8))`.

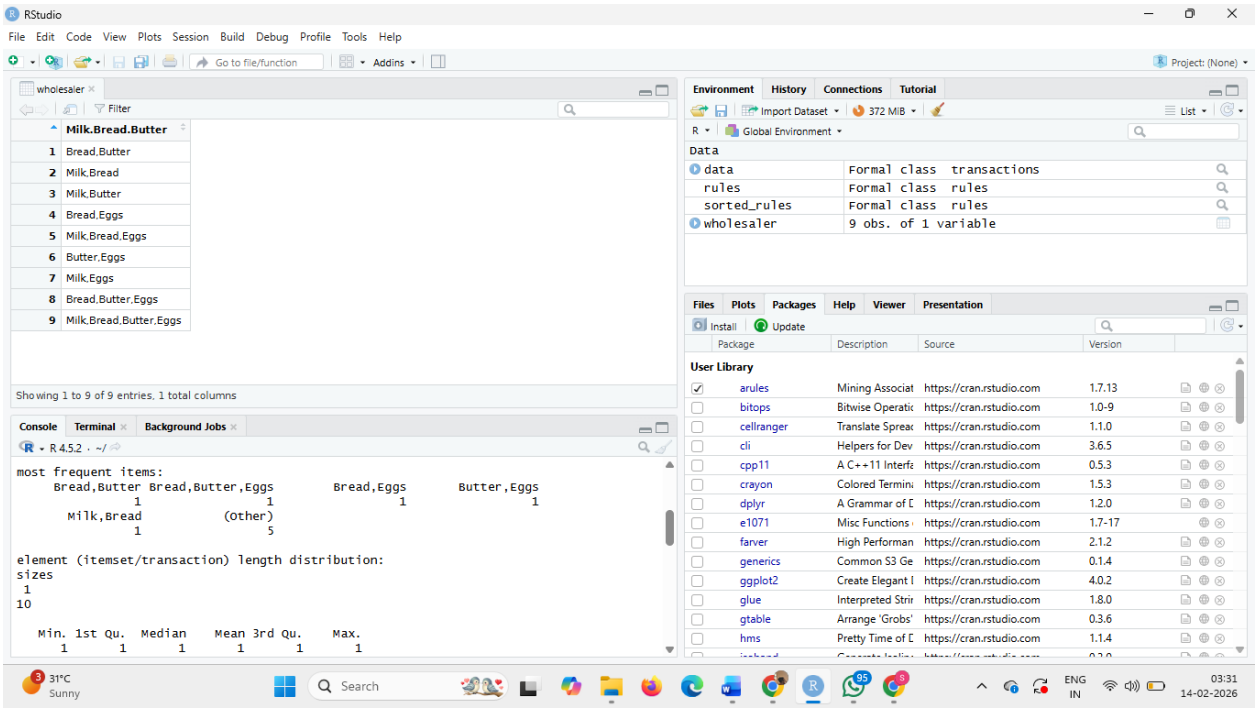
Step 5: View the top rules using `inspect (head (rules, 10))` and stop the program.

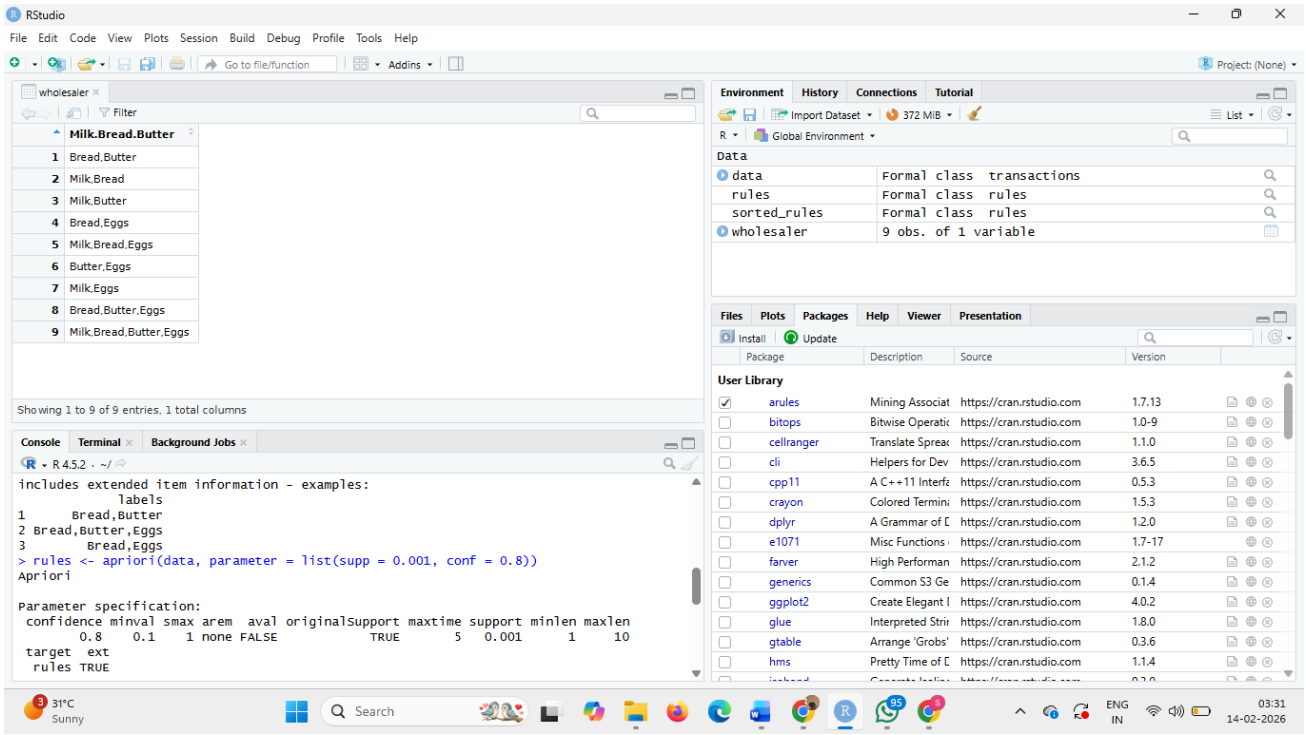
SOURCE CODE:

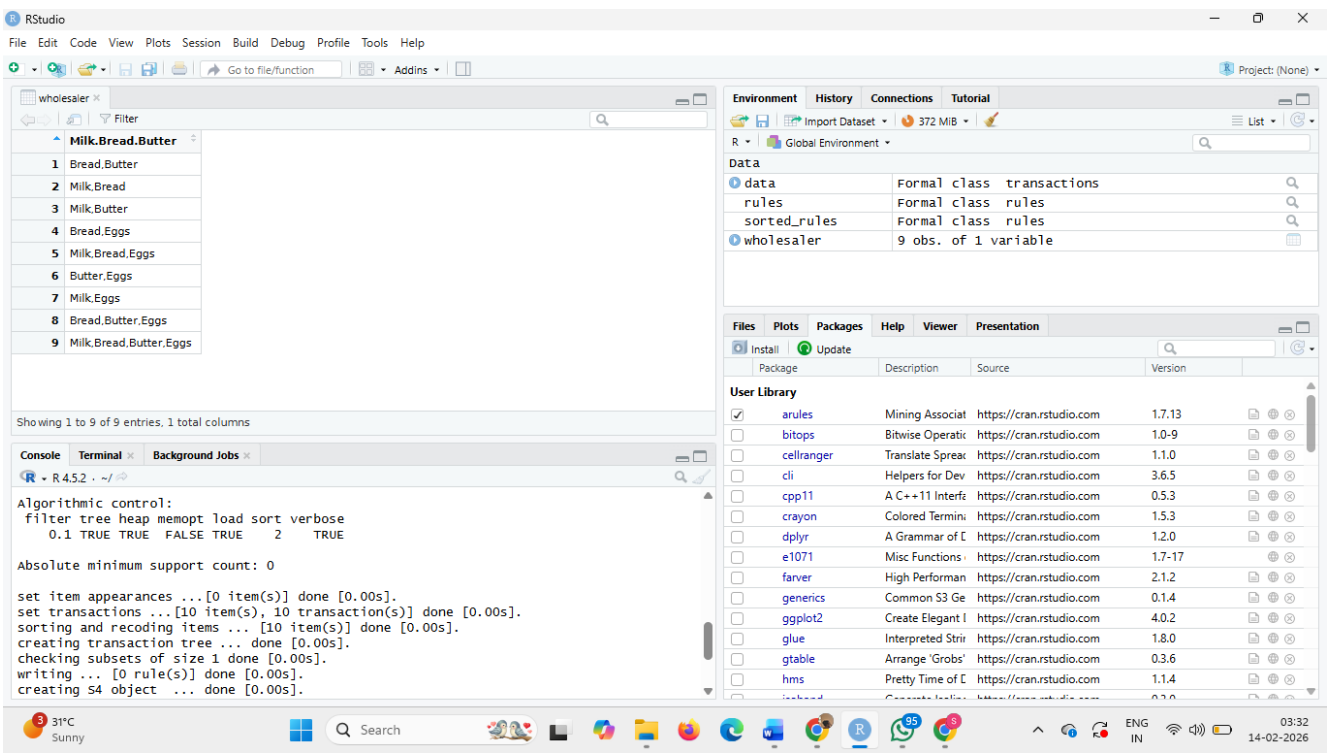
```
library(arules)
data <- read.transactions(file.choose(), format = "basket", sep = ",")
summary(data)
rules <- apriori(data, parameter = list(supp = 0.001, conf = 0.8))
length(rules)
inspect(head(rules, 10))
inspect(rules)
sorted_rules <- sort(rules, by = "confidence", decreasing = TRUE)
inspect(head(sorted_rules, 10))
```

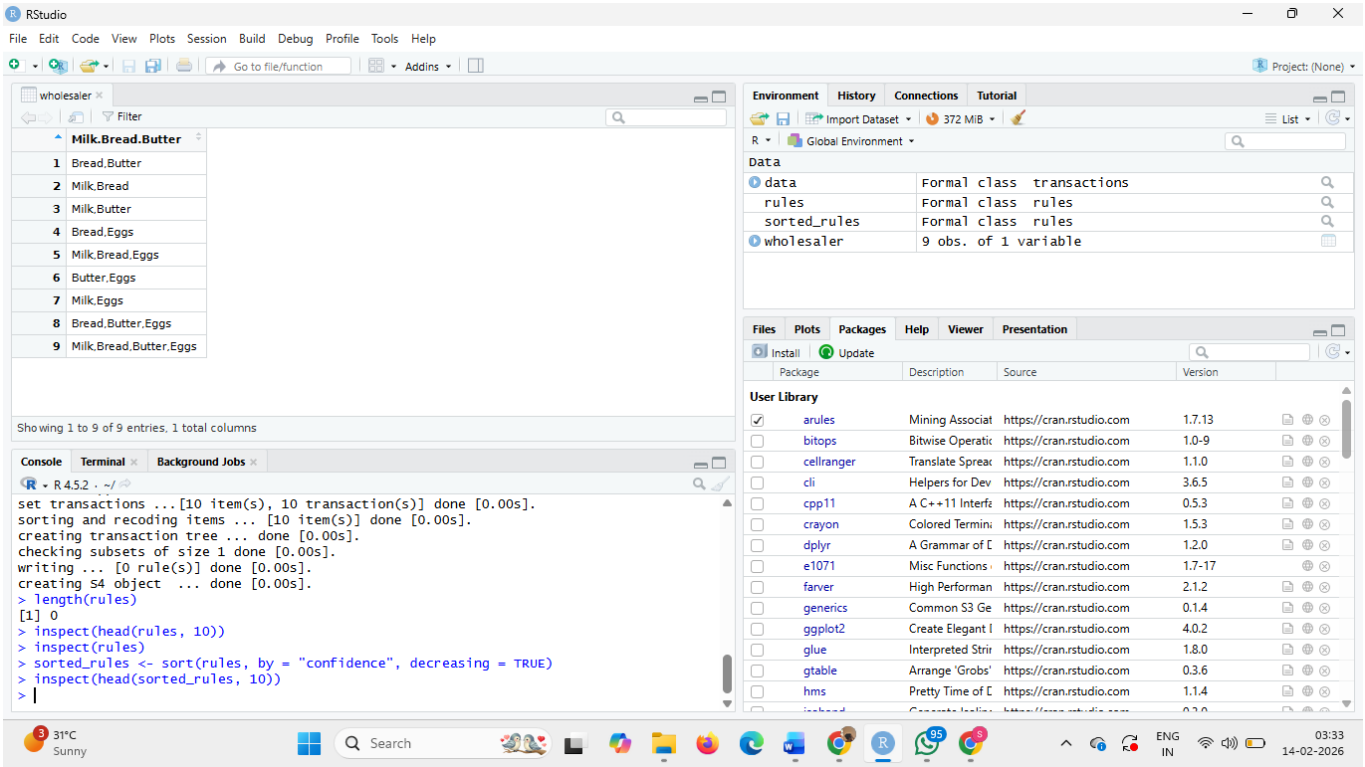
OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
wholesaler
Filter
Milk.Bread.Butter
1 Bread,Butter
2 Milk,Bread
3 Milk,Butter
4 Bread,Eggs
5 Milk,Bread,Eggs
6 Butter,Eggs
7 Milk,Eggs
8 Bread,Butter,Eggs
9 Milk,Bread,Butter,Eggs
Showing 1 to 9 of 9 entries, 1 total columns
Environment History Connections Tutorial
Data
data Formal class transactions
rules Formal class rules
sorted_rules Formal class rules
wholesaler 9 obs. of 1 variable
Console Terminal Background Jobs
> library(arules)
Attaching package: 'arules'
The following objects are masked from 'package:base':
abbreviate, write
> data <- read.transactions(file.choose(), format = "basket", sep = ",")
> summary(data)
transactions as itemMatrix in sparse format with
10 rows (elements/itemsets/transactions) and
10 columns (items) and a density of 0.1


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
wholesaler
Milk.Bread.Butter
1 Bread,Butter
2 Milk,Bread
3 Milk,Butter
4 Bread,Eggs
5 Milk,Bread,Eggs
6 Butter,Eggs
7 Milk,Eggs
8 Bread,Butter,Eggs
9 Milk,Bread,Butter,Eggs
Showing 1 to 9 of 9 entries, 1 total columns
Console Terminal Background Jobs
most frequent items:
Bread,Butter Bread,Butter,Eggs Bread,Eggs Butter,Eggs
1 1 1 1
Milk,Bread (Other)
1 5
element (itemset/transaction) length distribution:
sizes
1
10
Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
wholesaler
Milk.Bread.Butter
1 Bread,Butter
2 Milk,Bread
3 Milk,Butter
4 Bread,Eggs
5 Milk,Bread,Eggs
6 Butter,Eggs
7 Milk,Eggs
8 Bread,Butter,Eggs
9 Milk,Bread,Butter,Eggs
Showing 1 to 9 of 9 entries, 1 total columns
Environment History Connections Tutorial
Data
data Formal class transactions
rules Formal class rules
sorted_rules Formal class rules
wholesaler 9 obs. of 1 variable
Console Terminal Background Jobs
R 4.5.2 · ~/
includes extended item information - examples:
labels
1 Bread,Butter
2 Bread,Butter,Eggs
3 Bread,Eggs
> rules <- apriori(data, parameter = list(supp = 0.001, conf = 0.8))
Apriori
Parameter specification:
confidence minval smax arem aval originalSupport maxtime support minlen maxlen
0.8 0.1 1 none FALSE TRUE 5 0.001 1 10
target ext
rules TRUE


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
wholesaler
Milk.Bread.Butter
1 Bread,Butter
2 Milk,Bread
3 Milk,Butter
4 Bread,Eggs
5 Milk,Bread,Eggs
6 Butter,Eggs
7 Milk,Eggs
8 Bread,Butter,Eggs
9 Milk,Bread,Butter,Eggs
Showing 1 to 9 of 9 entries, 1 total columns
Environment History Connections Tutorial
Data
data Formal class transactions
rules Formal class rules
sorted_rules Formal class rules
wholesaler 9 obs. of 1 variable
Console Terminal Background Jobs
R 4.5.2 · ~/
Algorithmic control:
filter tree heap memopt load sort verbose
0.1 TRUE TRUE FALSE TRUE 2 TRUE
Absolute minimum support count: 0
set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[10 item(s), 10 transaction(s)] done [0.00s].
sorting and recoding items ... [10 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 done [0.00s].
writing ... [0 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].



RESULT:

The above program is executed successfully.